Splitting the Stack: Generative AI and the Cross-Country Division of Technical Work

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Abstract

American multinationals split their technical work across countries, building models at home and testing and running software abroad. I follow about 4,500 U.S. multinationals across the United States, India, the United Kingdom, Mexico and the Philippines, read each job posting into one of seven kinds of technical work, and interpret placements through a multi-destination model of trade in tasks. Destinations do not rank tasks alike: India specializes in technical application and testing, the United Kingdom and the Philippines in business-facing application and governance, and the most doctoral-intensive jobs are 30 to 81 percent less likely to be placed abroad. Over a decade India gained every kind of work, yet its gap in model building never closed. Generative AI did not shift this division of labor; it changed hiring at home, where junior software-developer postings fell 19 percent relative to senior ones in 2023 and recovered by 2024.

Keywords: offshoring, trade in tasks, generative AI, entry-level employment, multinational firms, high-skilled immigration.

JEL codes: F16, F23, J23, J24, O33.

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1 Introduction

When a bank in New Jersey posts a job for a data scientist, the work may be done in New Jersey or in India. In 2024 one bank posted two jobs under exactly that title, in the same job family. The one in Jersey City asked the worker to build deep-learning models in PyTorch; the one in India asked for SQL queries and Tableau dashboards (Table 1). Splitting a single title across countries is how large American firms now organize technical work. They do not offshore occupations; they split each occupation's work across countries, keeping some of it at home and sending the rest abroad. In the firms I study, India's share of their combined U.S. and Indian jobs in these kinds of technical work rose from 15 to 35 percent between 2019 and 2025, partly because the data provider's coverage of India grew.

Generative AI arrived in the middle of this division of labor, and it could move it in two opposite directions. If it makes remote work easier to coordinate and check, it should send more technical work abroad. If it instead does the codified work that firms have already sent abroad, it should reduce hiring wherever that work is done, without moving any of it. Which happened matters for whose entry-level job is at risk, for whether restricting offshoring or high-skilled immigration protects U.S. workers, and for where the next generation of senior engineers is trained.

Did generative AI move technical work out of the United States? To answer, I measure where 4,549 U.S. multinationals place seven kinds of technical work—from building models to running help desks—in the United States, India, the United Kingdom, Mexico and the Philippines, using 56 million job postings collected from the firms' own career sites. I read the kind of work from each posting's text with definitions that contain no AI vocabulary, so that a posting's label does not change when its employer starts to mention AI. A multi-destination version of the Grossman–Rossi-Hansberg model of trade in tasks turns the firms' placement of each kind of work into an equation I can estimate within firms, and it implies that a technology which saves the same labor wherever a task is done moves no tasks. I then test, with designs written down before estimation, for a break in task location at ChatGPT's release and for changes in the experience mix of hiring in each country. The answer is no. No kind of technical work shows a break in India's share at the release, measured against routine IT support, while U.S. junior software-developer postings fell 19 percent relative to senior ones in 2023 and recovered by 2024: generative AI changed, for about a year, whom these firms hired at home, not where they do their technical work.

The question sits where two literatures meet, and neither can answer it on its own. Trade-in-tasks theory begins with Grossman and Rossi-Hansberg (2008), in which a fall in the cost of distance sends tasks abroad and can raise home wages through a productivity

effect. The model ranks tasks by a single cost of remote performance, so every destination should receive the same tasks first. Empirical work could not test that ranking, because it measured offshorability by occupation (Blinder and Krueger 2013) and estimated effects by industry, occupation or imported input; Liu and Trefler (2011), for example, find no significant earnings losses for U.S. white-collar workers exposed to service imports from China and India once worker sorting is accounted for, but they cannot see which tasks moved. Assignment models add that skill-intensive tasks should go to skill-abundant countries (Costinot 2009), a prediction that has not been tested inside the firm.

The second literature asks what generative AI does to early-career workers, and it finds effects. Eloundou et al. (2024) measure which tasks language models can perform. Westby and Modestino (2026) show that after ChatGPT's release junior U.S. software-developer vacancies fell relative to senior ones and to other technical occupations; payroll records (Brynjolfsson et al. 2025) and résumés (Hosseini and Lichtinger 2025) point the same way, and Chandar and Klein Teeselink (2026) extend the comparison to 41 countries. Friebel et al. (2026) show in theory why a junior hiring freeze after a productivity shock can be temporary.

These studies compare occupations or countries. None observes where the same firm performs each task, before and after the technology arrives, so none can ask whether the technology moved work rather than removed it. The trade literature has the right question about location but not the right unit; the AI literature has a credible design for entry-level hiring but sees only one side of the firm.

This paper makes four contributions. The first is to internal validity. Every comparison is made within firm, country and period, so wages, LinkUp's coverage of each country and the firm's overall demand are absorbed by fixed effects, and the volume of postings never enters as an outcome. The relocation test compares the break at ChatGPT's release with breaks at eleven placebo dates, which show what relocation looks like in these data: none of the six breaks at the release is unusual (rank p-values 0.08 to 0.92), and in every category the 95 percent interval excludes a break as large as the largest one of the remote-work period (+0.68 log points, in 2020–21). The test detects relocation of technical work relative to routine support; relocation that moved every kind of work abroad in equal proportion would be indistinguishable from the growth of the data provider's coverage. Within the same firms I reproduce the U.S. estimate of Westby and Modestino (2026) (-0.155 against their -0.136), show that a junior decline of 0.215 log points (s.e. 0.063) is concentrated in 2023 and gone by 2024, and find no counterpart in the same firms' Indian operations. I report every pre-specified design, with multiple-testing adjustments, including those that fail.

The second contribution is to external validity: the paper follows the work abroad.

Occupation codes, the usual unit, cover almost none of the British, Mexican or Philippine postings and lump together much of what differs between countries; one code, Software Developers, holds 39 percent of India's technical postings. Reading tasks from text, I find that destinations do not rank tasks alike, as the single-ranking model requires. India's ordering of the six categories other than routine support (the *focal* categories) is negatively correlated with the United Kingdom's (-0.31) and the Philippines' (-0.43): India specializes in technical application and testing, the United Kingdom and the Philippines in business-facing application and governance. At the top of the skill distribution there is a doctoral margin: titles for which more than 30 percent of U.S. postings ask for a PhD are 30 to 81 percent less likely to be placed abroad. Between 2016 and 2025 India gained every category of technical work, yet model building trailed application work by 0.42 log points at both ends of the decade.

The third contribution is to measurement. I read technical work from posting text in two ways, both built to be compared across countries and years. A full-text classifier assigns each U.S. and Indian posting since 2016 to one of seven categories whose definitions contain no AI vocabulary, so a posting's label does not change when its employer starts to mention AI; on postings whose titles leave no doubt, it is right 90.5 percent of the time with the title and 73.9 percent without it. A topic model, labelled group by group rather than word by word, fixes each job title's category from its U.S. postings and so labels the same title identically in five countries. Both measures vary within occupations—software developers are 48.5 percent technical application in the United States and 65.4 percent in India—which is what lets the paper see the division of labor that occupation codes hide. Posting text has been used to track how the content of jobs changes (Hershbein and Kahn 2018; Deming and Kahn 2018; Acemoglu et al. 2022); here it also locates that content within the multinational.

The fourth contribution is to theory. I extend Grossman and Rossi-Hansberg (2008) to several destinations that price task attributes differently and differ in graduate supply. The extension yields a gravity equation for jobs that can be estimated with firm fixed effects, a sorting result in the spirit of Costinot (2009), a neutrality result that separates relocation from location-neutral labor saving, and the home share of jobs as a sufficient statistic for the productivity effect on U.S. pay. With one white-collar labor market the model implies that pay in offshored tasks keeps pace with other white-collar pay; the alternative of segmented markets, in which released workers depress their own group's wage, implies that it falls behind. The data side with the model: as India's share of these tasks roughly tripled between 2019 and 2025, U.S. pay in them rose by about 25 percent, as much as other white-collar pay.

The remainder of this paper is organized as follows. Section 2 describes the setting: the

four destinations, why occupation codes cannot locate tasks, and the events that coincided with ChatGPT's release. Section 3 develops the model and derives its testable predictions. Section 4 presents the data and descriptive statistics, and Section 5 builds and validates the two measures of tasks read from posting text. Section 6 lays out the estimating equations and discusses what they identify. Section 7 presents the results, from where firms place each task to the relocation test and entry-level hiring, followed by robustness checks, heterogeneity and mechanisms, including U.S. pay. Section 8 concludes with implications for policy and for future research.

2 Background

2.1 U.S. multinationals and their four destinations

The firms studied here are the 4,549 U.S. multinationals that posted technical jobs in both the United States and India between 2019 and 2025, excluding KPMG, Deloitte and Infosys. Each of those three posts several Indian jobs for every U.S. job, in proportions unlike any other firm, and they are the subject of a companion paper. Besides India, I follow the firms' technical jobs into the United Kingdom, Mexico and the Philippines, the destinations, among the firms' largest, whose job titles are written in English and can be classified on identical terms. They differ in ways the model of Section 3 makes central. India and the Philippines are distant, low-wage and share few working hours with the United States. Mexico is low-wage, near and shares U.S. time zones. The United Kingdom pays wages close to American ones and serves a large market of its own.

The destinations also differ in the graduate labor they have. The companion paper Pandit and Nambiar (2026) documents it from worker profiles: 15.5 percent of U.S. and 18.3 percent of British AI workers hold a PhD; doctorates are less common in Mexico and rarer still in India (1.8 percent) and the Philippines (2 percent or less), and 62.5 percent of India's AI workers stopped at a bachelor's degree. India keeps only 37 percent of its graduate-educated AI workers, and half of them live in the United States. The model uses only the ordering of the destinations, and none of that paper's data enters the estimation.

2.2 Why occupation codes cannot locate tasks

The natural unit for a study of offshoring would be the occupation. Every U.S. and Indian posting in the data carries an O*NET occupation code matched from its title. For this question the code fails in three ways. It is coarse: Software Developers (15-1252) contains 39.2 percent of India's technical postings and 25.4 percent of the United States', and India's

distribution across codes is much more concentrated (Herfindahl index 0.185 against 0.112). It is static: the code is assigned from the title, and the same title can describe very different work. And it is unavailable outside the United States and India, where fewer than 0.01 percent of the firms' British, Mexican or Philippine postings carry a code.

Table 1 gives the case from the introduction. In 2024 JPMorgan Chase advertised two "Data Scientist" jobs, both in the job family it calls Predictive Science: a vice-president role in Jersey City and a senior associate role in India. An occupation code records the two as the same job, and so does the title. The example should not be over-read. Across all "data scientist" postings in 2024–25, about half are classified as Building in each country (48 percent in the United States, 51 percent in India), and the shares are nearly identical within the 359 firms that post the title in both. The point is not that the title means one thing in each country. It is that a single title, at a single firm in a single year, covers both kinds of work, and only the text can tell them apart.

Table 1: One title, two jobs: "Data Scientist" at JPMorgan Chase, 2024

United States — "Data Scientist – Vice President," Jersey City, NJ Classified: model building (Building)	*India* — "Data Scientist Associate Sr" Classified: functional application (Application)
"Develop state-of-the art machine learning models to solve real-world problems and apply it to tasks such as NLP, personalization, or recommendation systems."	"Writing SQL Codes/Stored procedures/Python code capturing all the logics to identify the exceptions and testing data, developing Tableau dashboards."
"Develop end-to-end ML/AutoML/AutoNLP pipelines and operationalize the end-to-end orchestration of the ML models."	"Ensure codes are designed efficiently to maximize performance and consistency of reporting environment. Enhance ad-hoc reporting capabilities."
"Solid background in NLP, Generative AI … familiar with large language models." "Extensive experience with machine learning and deep learning toolkits (e.g.: Transformers, Hugging Face, TensorFlow, PyTorch …)."	"Hands-on experience with standard data analytics tools and technologies (SQL, Tableau, Alteryx, Python & Snowflake)."
"Published research in areas of Machine Learning, Deep Learning or Reinforcement Learning at a major conference or journal" (preferred).	"Interact with the stakeholders to understand business needs … Providing regular updates to key stakeholders and senior leaders."

Notes: Verbatim excerpts from the two postings as published on the employer's careers site and collected by LinkUp (U.S. job ID 210494314, posted February 28, 2024; India job ID 210522349, posted June 14, 2024). Both are listed by the employer under the job family "Predictive Science." Selected for clarity, not as representative of a country difference (see text).

2.3 What else happened around November 2022

ChatGPT's release on November 30, 2022 did not happen in isolation. It coincided with higher interest rates, technology-sector layoffs and a slowdown of fresher hiring in Indian technical work, and it followed by two years the shift to remote work, which also changed the cost of working away from the office. A test for relocation must therefore separate a break at the release from these other changes, and it must be able to detect relocation when relocation occurs. Section 6.2 explains how the designs do both.

3 Theoretical Framework

I adapt the model of Grossman and Rossi-Hansberg (2008), henceforth GRH, to an environment that matches the data: firms place individual jobs rather than whole tasks, they can place them in several destinations, the cost of doing a task away from headquarters has several sources that destinations price differently, and tasks differ in the graduate labor they need while destinations differ in the graduate labor they have. The theory of change the paper tests is the following. Generative AI can reach the allocation of tasks in two ways: by lowering the cost of performing a task far from headquarters, which moves tasks abroad, or by saving labor wherever a task is performed, which changes hiring in every country and moves no tasks. Proofs are in Appendix I. The testable predictions are labelled P1 to P7 where they arise and collected in Section 3.8.

3.1 What the extension adds

GRH model offshoring as trade in tasks and show that a fall in the cost of offshoring can *raise* the wage of the home workers who remain, because it acts like factor-augmenting technical progress. To make that point they assume one foreign country, a single index $t(z)$ that ranks tasks by their cost of remote performance, and all-or-nothing allocation of each task; they distinguish low- and high-skill tasks, but tasks do not differ continuously in the graduate labor they need, nor destinations in the graduate labor they have. Table 2 sets their environment beside the one used here.

3.2 Environment

A U.S.-headquartered firm i produces output by combining tasks $z \in \mathcal{Z}$. In year t it has N_{izt} jobs of task z to fill, and it places each job j in one country $c \in \mathcal{C} = \{\text{US}\} \cup \mathcal{D}$, with destinations $\mathcal{D} = \{\text{IN}, \text{PH}, \text{MX}, \text{UK}\}$. Each country has graduate (H) and non-graduate (L)

Table 2: The GRH environment and the extension

	Grossman and Rossi-Hansberg (2008)	This paper
Unit of allocation	A task, all or nothing	A job; a task's jobs are split across countries by idiosyncratic cost draws
Destinations	One	Several, each with its own costs
Cost of remote performance	One index $t(z)$: all destinations rank tasks alike	$\log \tau_c(z) = \kappa_c + x_z'\delta_c$: several sources, priced differently by each destination
Labor	Two skill types, each with its own tasks	Continuous graduate intensity s_z; destinations differ in graduate supply; graduates migrate
Technical change	A fall in the cost of offshoring	A fall in location costs τ, or a fall in labor requirements a common to all locations
Home wages	Productivity, relative-price and labor-supply effects	Productivity effect with the home share as a sufficient statistic; with one white-collar labor market, offshoring raises the common wage
Empirical content	Comparative statics	A gravity equation for jobs with firm fixed effects, estimable from postings

technical labor with wages w_c^H and w_c^L; $\rho_c \equiv \log(w_c^H/w_c^L)$ is the log graduate premium. Task z requires $a(z)$ units of labor per job, a share $s_z \in [0,1]$ of it graduate labor. The cost of job j of task z in country c is

$$C_{ijzc} \;=\; \frac{a(z)\,\tau_c(z)\left(w_c^H\right)^{s_z}\left(w_c^L\right)^{1-s_z}\phi_{ic}}{\varepsilon_{ijc}}, \tag{1}$$

where $\tau_c(z) \geq 1$ is the cost of performing z in c relative to the United States ($\tau_{US}(z) = 1$), ϕ_{ic} collects everything specific to the firm in the country (its pay premium, the cost of running an office there), and ε_{ijc} is a job-specific efficiency draw that stands for what the researcher does not see: a team's needs, spare capacity, a manager's preference. Each job is placed where its cost is lowest.

Assumption 1 (efficiency draws). ε_{ijc} is independent across jobs and countries and distributed Fréchet, $\Pr(\varepsilon \leq e) = \exp(-e^{-\theta})$, $\theta > 0$.

Assumption 2 (location costs). $\log \tau_c(z) = \kappa_c + x_z'\delta_c$, where x_z is a vector of task characteristics and $\delta_{US} = 0$, $\kappa_{US} = 0$. In the main estimates x_z is the task's category; Appendix G uses measured requirements.

Assumption 3 (task skill intensity). s_z does not vary across countries.

Assumption 4 (graduate premia). In each country the relative demand for

graduate technical labor has constant elasticity $\sigma > 0$: $H_c/L_c = B_c\, e^{-\sigma \rho_c}$, so $\rho_c = \sigma^{-1}\big(\log B_c - \log(H_c/L_c)\big)$. The firms take ρ_c as given.

Assumption 1 makes the allocation of a task's jobs a discrete-choice problem with the extreme-value structure of Eaton and Kortum (2002); θ, the placement elasticity, is the analogue of the trade elasticity. Assumption 2 is essential: it lets each destination price each kind of task differently, so that work close to the business can be cheap to place in London and costly in Bangalore, while engineering work can be the reverse. Assumption 3 is simplifying; if firms redesign a task to use less graduate labor where graduates are scarce, the skill coefficients below are attenuated. Assumption 4 is the simplest link from a country's graduate supply to its premium; B_c collects the demand for graduate labor from all employers.

With one destination F, homogeneous labor ($s_z = 0$), $\tau_F(z) = b\, t(z)$ and $\theta \to \infty$, every job of task z goes abroad if and only if

$$w_F\, b\, t(z) < w_{US}, \tag{2}$$

which is the GRH cutoff rule: tasks below a threshold are offshored, and a fall in b raises the threshold. The finite-θ model replaces the cutoff with shares, which is what makes the same title appear in several countries at once.

3.3 The allocation of tasks across countries

Lemma 1 (job shares). Under Assumption 1 the share of firm i's jobs of task z placed in country c is

$$\pi_{izc} \;=\; \frac{\big(\tau_c(z)\,(w_c^H)^{s_z}(w_c^L)^{1-s_z}\,\phi_{ic}\big)^{-\theta}}{\sum_{c'}\big(\tau_{c'}(z)\,(w_{c'}^H)^{s_z}(w_{c'}^L)^{1-s_z}\,\phi_{ic'}\big)^{-\theta}}, \tag{3}$$

and under Assumptions 1–3 the log odds of destination d against the United States are

$$\frac{1}{\theta}\log\frac{\pi_{izd}}{\pi_{iz,US}} \;=\; \underbrace{\omega_{id} - \kappa_d}_{\text{common to the firm's tasks}} \;-\; \underbrace{x_z'\delta_d}_{\text{location cost}} \;-\; \underbrace{s_z\,(\rho_d - \rho_{US})}_{\text{skill cost}}, \tag{4}$$

where $\omega_{id} \equiv \log\big(w_{US}^L \phi_{i,US}/(w_d^L \phi_{id})\big)$ is the wage wedge.

Two features of Lemma 1 organize everything that follows. First, the labor requirement $a(z)$ is common to all countries and cancels from Equation (3): location depends on how much *more* a task costs away from headquarters, not on how much labor it needs. Second,

the wage wedge ω_{id} and the general cost of operating abroad κ_d are common to all of the firm's tasks in a destination, so they govern how much the firm offshores there but not which tasks go. Which tasks go is governed by the location-cost term and the skill term. The Fréchet structure also implies independence of irrelevant alternatives across destinations, a restriction that matters for counterfactuals that reallocate jobs among destinations, not for the estimates below, which compare each destination with the United States.

The data observe only a fraction η_{ict} of the firm's postings in a country and year (LinkUp's coverage), so Lemma 1 is taken to data as a count model.

Proposition 1 (Gravity equation for jobs). *Under Assumptions 1–3, and if coverage η_{ict} does not vary across the firm's tasks within a country and year, the expected number of observed jobs of task z that firm i places in country c in year t is*

$$\mathbb{E}[V_{iczt}] = \exp\big(\alpha_{izt} + \gamma_{ict} + x_z'\beta_c^{\tau} + s_z\,\beta_{ct}^{s}\big), \qquad \beta_c^{\tau} = -\theta\,\delta_c, \quad \beta_{ct}^{s} = -\theta\,(\rho_{ct} - \rho_{US,t}), \tag{5}$$

where α_{izt} varies only by firm, task and year and γ_{ict} only by firm, country and year.

In words: the number of a firm's jobs of a given kind in a given country depends on how much of that work the firm needs, on how much it hires in that country overall, and on two prices—the destination's price of the task's characteristics and of its graduate intensity. The fixed effects carry everything else, so none of it has to be measured. Poisson pseudo-maximum likelihood with these fixed effects is consistent for the two prices because, conditional on the firm-task-year total, the counts are multinomial with the shares of Lemma 1 (Santos Silva and Tenreyro 2006).

3.4 Comparative advantage in tasks

Equation (4) contains a two-factor comparative-advantage structure inside the firm. Every destination saves the firm money on the non-graduate part of a task; only a destination where graduates are relatively plentiful also saves on the graduate part.

Proposition 2 (Skill sorting). *Hold graduate demand fixed ($B_d = B_{US}$). (i) The odds $\pi_{izd}/\pi_{iz,US}$ fall with skill intensity s_z if and only if graduates are relatively scarcer in d than in the United States, $H_d/L_d < H_{US}/L_{US}$. (ii) The skill coefficients are ordered by relative graduate supply: $\beta_d^s > \beta_{d'}^s$ if and only if $H_d/L_d > H_{d'}/L_{d'}$. Equivalently, the odds of placing a task in a destination are log-supermodular in the task's graduate intensity and the destination's relative graduate supply.*

The result is in the spirit of Costinot (2009) and Costinot and Vogel (2010): when productivity is log-supermodular in a task characteristic and a country characteristic, the

more skill-intensive tasks are assigned to the more skill-abundant countries. Given the graduate supplies of Section 2, the prediction is $\beta^s_{UK} \gtrsim 0 > \beta^s_{MX} > \beta^s_{IN} \gtrsim \beta^s_{PH}$ (P4). The demand shifter works against the prediction: a country with few research employers has a lower premium for given supply, which makes graduate-intensive tasks cheaper to place there. A negative coefficient for India is therefore evidence that scarce supply dominates, not an artefact of thin demand.

Graduate supply in a destination is itself endogenous to migration. Let $\bar{H}_d$ be the graduates trained in d and m_d the share who work in the United States, so $H_d = (1 - m_d)\bar{H}_d$ and $H_{US} = \bar{H}_{US} + \sum_d m_d \bar{H}_d$.

Proposition 3 (Migration and offshoring of graduate-intensive tasks). *Holding $\bar{H}$, L and B fixed, a rise in m_d lowers β^s_d, and lowers $\beta^s_{d'}$ for every other destination d' through the rise in H_{US}.*

Emigration makes graduate labor scarcer in the origin and more plentiful at home, steepening the gradient from both sides. Hiring the migrant and offshoring the task are therefore substitutes for graduate-intensive work, so restricting high-skilled immigration moves graduate-intensive tasks abroad, as Glennon (2024) finds for the H-1B program. I do not observe migration by firm; the proposition organizes facts rather than being tested.

Model building is both tied to headquarters and graduate-intensive. Differencing Equation (4) between a Building task and an Application task in the same destination removes ω_{id} and κ_d and leaves their difference in location attributes, their difference in skill intensity times the destination's premium gap, and a Building-specific location cost δ^B_{dt} that the measured attributes miss (proprietary data, compute, proximity to research teams; Equation (16) in the appendix).

Proposition 4 (What stays home, and when it moves). *(i) If Building is more graduate-intensive than Application, the skill component of the Building–Application gap in offshore odds is most negative where graduates are scarcest and zero where relative supply equals that of the United States. (ii) Its skill component disappears once destination × skill intensity is controlled. (iii) The gap changes over time only through the Building-specific location cost or the destination's relative graduate supply. In particular, a fall in location costs common to every task above routine support, $\Delta \log \tau_d(z) = -\xi_{dt}$ for all such z, raises the destination's odds for every such task relative to support by $\theta\xi_{dt}$ and leaves the gap unchanged.*

Part (iii) gives a precise meaning to a destination *learning its way up the stack*: it gains every category above support at once, and the gap at the top closes only if the destination's graduate supply rises or the specific cost of model building falls (P3, P5).

3.5 Technical change: relocation versus location-neutral labor saving

Let $a^J(z) \leq a(z)$ be the junior labor in a job of task z, and ι_z the junior share of the job's wage bill. Writing $\hat{x} \equiv x'/x$ for the proportional change in any variable, as in Dekle et al. (2007), a technology can act on Equation (1) in two ways: through location costs $\hat{\tau}_c(z)$, as cheaper remote coordination does, or through the labor requirement, as a coding assistant that saves the same junior labor wherever the task is done does.

> *Assumption 7 (common junior cost share).* ι_z is the same in every country, so a given proportional saving of junior labor lowers the cost of a job of task z by the same proportion, $\psi_z \iota_z$, wherever it is performed.

Proposition 5 (Two channels for generative AI: relocation and location-neutral labor saving). *(i) If AI lowers the cost of performing technical tasks away from headquarters relative to routine support, $\Delta \log \tau_d(z) = -\xi_{dt}$ for every task z above support, then the odds of placing each such task in d rise relative to support by $\theta \xi_{dt}$ and, holding N_{izt} fixed, junior hiring in d rises relative to the United States. A fall common to every task, support included, changes only κ_{dt}. (ii) Under Assumption 7, if AI lowers the junior labor of task z, $a^J(z)$, by a factor $1 - \psi_z$ in every country, then $\hat{\pi}_{izc} = 1$ for all c: no job moves, and junior hiring in country c falls by*

$$\Delta \log J_c \;\approx\; -\sum_z \nu_{zc}\, \psi_z, \tag{6}$$

where ν_{zc} is task z's share of junior hiring in country c.

Part (ii) is a neutrality result: a labor saving common to all locations is, like $a(z)$ itself, absent from Equation (3). Without Assumption 7 it holds only approximately: if juniors are a larger share of a task's cost in one country, the same saving lowers that country's cost more and draws the task toward it. The proposition yields the cleanest test of the two channels because it concerns the allocation, not employment levels: under (ii) a destination's share of each task shows no break when the technology arrives, under (i) its share of every task above support jumps relative to support. Part (i) also says what the test cannot see. A fall in location costs common to every task changes only κ_{dt}, which the estimating equations absorb together with the data provider's coverage. Under (ii) the incidence of the technology across countries is set by the allocation. Offshoring selection sends the codified, low-s_z slice of each occupation abroad, so the junior decline is broad offshore and concentrated in the most exposed occupation at home; an occupation-level contrast therefore detects the effect in the United States more easily than in India (P6).

3.6 Home wages

Group tasks into Building and Application, and let w_g be the wage of the U.S. workers who perform group g.

> *Assumption 5 (small open economy).* The price p_g of the output of group g is set in world markets and firms earn zero profits. Within a group, tasks enter with Cobb–Douglas cost shares λ_z, $\sum_{z \in g} \lambda_z = 1$.

Proposition 6 (Productivity effect). *Under Assumptions 1–5, with labor requirements fixed,*

$$\Delta \log w_g \;=\; -\frac{1}{\theta} \sum_{z \in g} \lambda_z \, \Delta \log \pi_{z,US} \;+\; \Delta \log p_g. \tag{7}$$

With output prices fixed, a fall in the share of the group's jobs performed at home raises the wage of the U.S. workers who remain, by $1/\theta$ times the fall in the log home share. The home share is a sufficient statistic, as the domestic expenditure share is for the gains from trade in Arkolakis et al. (2012). Relative to *task autarky*, in which every job is performed at home, the gain at fixed prices is $-\frac{1}{\theta} \sum_z \lambda_z \log \pi_{z,US}$. Using the statistic requires θ, which the gravity estimates below do not identify. Under Assumption 5 output prices pin down the group's wage, so GRH's labor-supply effect, which arises when workers released from offshored tasks must be absorbed by a market with fixed supply, cannot operate; their relative-price effect enters through $\Delta \log p_g$.

Engineers, analysts and managers move between Building and Application work and between them and other white-collar work. Closing the model with a common wage makes the productivity effect economy-wide.

> *Assumption 6 (one white-collar labor market).* U.S. white-collar workers are mobile across task groups and earn a common wage w_W. The groups combine into traded white-collar services with Cobb–Douglas cost shares μ_g, $\sum_g \mu_g = 1$, whose price p is set in world markets.

Proposition 7 (Offshoring and the U.S. white-collar wage). *Under Assumptions 1–3, 5 and 6,*

$$\Delta \log w_W \;=\; \frac{1}{\theta} \sum_g \mu_g \Big(-\sum_{z \in g} \lambda_z \, \Delta \log \pi_{z,US} \Big) \;+\; \Delta \log p. \tag{8}$$

(i) With p fixed, a fall in home shares caused by lower foreign costs raises the wage of every U.S. white-collar worker, including those in groups whose home share did not fall. (ii) Pay in different task groups moves together. (iii) The wage falls only if offshoring lowers the

world price of white-collar services by more than $1/\theta$ times the cost-weighted fall in home shares.

A Building engineer in New Jersey gains from the offshoring of testing work in Bangalore for the same reason a worker gains from technical progress in a complementary task. Part (ii) predicts equal pay growth in Building and Application, and it implies that comparisons with other mobile white-collar work difference out the gain: such comparisons test Assumption 6 against segmented group labor markets, not the size of the productivity effect (P7′).

3.7 The experience ladder within each country

Appendix H extends the model to the shape of hiring within each country. Each operation is a three-layer knowledge hierarchy (Garicano and Rossi-Hansberg 2006) in which generative AI lowers both the number of juniors each mid-level engineer needs and the share of problems that must be escalated to seniors. Proposition 8 gives the condition under which the ladder becomes a diamond, with junior and senior shares both falling. Offshoring selection makes that condition more likely abroad, where junior work is codified and seniors resolve local exceptions, than at home, where seniors resolve frontier and doctoral problems (P6′).

3.8 Testable predictions

Each prediction is a restriction on a coefficient of Equation (5), on Equation (7) or (8), or on Propositions 5 and 8.

P1. Selective offshoring. The category prices $\beta^{\tau}_{d,k}$ differ from zero: low-wage destinations receive some categories more and others less than general tech.

P2. Destination-specific rankings. Destinations order the categories differently; under GRH's single ranking the orderings coincide.

P3. What stays home. The Building–Application gap is most negative where graduates are scarcest and shrinks when destination × skill intensity is controlled.

P4. Sorting on skill. $\beta^{s}_{d} < 0$ where graduates are scarce, ordered by graduate supply.

P5. The margin moves. A general rise of a destination through the stack lifts every category relative to support and leaves the gap at the top unchanged.

P6. Generative AI. Location-neutral labor saving moves no tasks: no break in a destination's share of any category; the junior decline appears in the most exposed occupation at home. Relocation would raise every share abroad.

P6′. The ladder. A temporary junior dip at home and a lasting shift toward the middle offshore (Proposition 8, with the dynamics of Friebel et al. (2026)).

P7. Group pay. U.S. pay in offshored groups keeps pace with other white-collar pay. The alternative of segmented group labor markets, in which released workers lower their own group's wage (GRH's labor-supply effect), predicts that it falls behind.

P7′. The white-collar wage. Pay in Building and Application rises alike.

Proposition 3 organizes facts and is not tested. The gravity estimates identify products of θ with attribute prices or premium gaps, not either alone; they rank and compare destinations but do not price τ or ρ in dollars. With four destinations the skill-supply prediction is a test of ordering, not an estimate of θ/σ.

4 Data and Descriptive Statistics

4.1 Source and sample

The data are job postings collected by LinkUp, which scrapes employers' own career sites rather than aggregating job boards. The text of each posting is therefore the employer's own: there are no aggregator duplicates, no recruiter rewrites and no re-posted listings from third parties. Every task measure in this paper is read from that text.

The firm population is the 4,552 multinationals that posted technical jobs in both the United States and India between 2019 and 2025; dropping KPMG, Deloitte and Infosys (Section 2) leaves 4,549, of which 4,541 post in at least one of the five countries in 2019–2025. I read every posting of these firms from LinkUp's raw global records in the five countries. Where a record appears in more than one of LinkUp's 291 raw shards, I keep the earliest. The extract contains 55,968,263 postings in 2019–2025: 45.4 million in the United States, 5.87 million in India, 3.43 million in the United Kingdom, 867,043 in Mexico and 441,002 in the Philippines.

LinkUp records a job once for every location listed on it, so a job advertised in five U.S. cities appears five times, and the number of copies differs by country and by task: a U.S. Building job appears 3.07 times on average and a U.S. Application job 2.46 times; in India the figures are 1.63 and 1.98. Counting records would overstate the U.S. share of Building work. The unit of analysis for the five-country results is therefore the *distinct job*: a unique combination of firm, normalized job title and creation date. Collapsing copies leaves 30.6 million distinct jobs across all titles.

The analysis samples are:

- *Five-country sample:* distinct jobs whose titles belong to the fixed title lookup of Section 5.2: 1,076,309 jobs in the six focal task categories and 72,358 in ordinary technical work, in 1.86 million firm × country × title × year cells.

- *U.S.–India full-text sample:* every technical posting of the firms in the two countries, 13.6 million in 2019–2025, of which 72 percent (9.8 million) have a full description, plus 2.1 million in 2016–2018; excluding KPMG, Deloitte, Infosys and postings outside the two countries leaves 11.1 million labelled postings, in 391,132 firm × country × category × year cells from 4,677 firms (the 2016–2018 extract adds firms that do not post in 2019–2025).

- *Levels sample:* U.S. and Indian technical postings with an O*NET code and a stated minimum experience, in firm × occupation group × level × month cells (3,190 U.S. and 2,155 Indian firms in the main window).

4.2 What the sample represents, and what is missing

The sample represents the technical hiring of large U.S. multinationals with operations in India, as advertised on their own sites. The destinations are those whose job titles are written in English, and the twenty firms in the sample whose product is AI place work differently from the rest (Appendix J). Three further gaps matter.

First, postings are not hires, and LinkUp's coverage is not constant. How many postings LinkUp records for a firm depends on how completely it scrapes the firm's career site in a given month, as well as on how much the firm is hiring, and LinkUp's coverage of India in particular expanded over the sample. I never use the volume of postings as an outcome. Every comparison is about the *composition* of a firm's hiring—which tasks it places in which country, and at what level of experience—with each firm's country-specific volume in each period absorbed by fixed effects.

Second, full descriptions exist only for the United States and India. That is why the five-country results rest on a title-based measure and the U.S.–India results on a full-text measure (Section 5).

Third, not every posting states an experience requirement: 60 percent of U.S. and 58 percent of Indian technical postings do, 48 percent of Philippine and 43 percent of Mexican postings, but only 17 percent of British postings, where stating years of experience is uncommon. Postings without a stated requirement drop out of the levels analysis; nothing is imputed.

4.3 Variables

Table 3 lists every variable the paper uses and how it is measured. The task measures, read in two ways from posting text, and skill intensity are the paper's own construction; Section 5 describes and validates them.

Table 3: Variable descriptions

Variable (unit)	Description
Outcomes	
Distinct jobs (count)	Unique firm × normalized title × creation date, counted by firm, country, title and year. Titles are lower-cased and every run of non-alphanumeric characters is replaced with a space. Five-country sample.
Postings (count)	LinkUp records, counted by firm, country, task category and year (stack) or quarter (relocation test). U.S.–India full-text sample.
Postings by level (count)	Postings with a stated minimum experience, counted by firm, country, occupation group, level and month. Levels sample.
U.S. pay (dollars per year, nominal)	BLS OEWS mean annual wage by occupation, weighted by each task group's U.S. occupation mix in LinkUp; BrightQuery industry and metropolitan cuts of OEWS (to May 2023); WageScape posted salaries on lookup titles (3.29 million postings, 13,378 titles).
Task characteristics	
Task category, full text (T1–T7)	Nearest of seven one-sentence category descriptions that contain no AI vocabulary, read from the full posting by a sentence-embedding model (Section 5.1). U.S. and Indian postings, 2016–2025.
Task category, title lookup	Category fixed for each job title from its U.S. postings: at least 30 postings and at least 60 percent with one topic-model label, 2019–2025 (Section 5.2). All five countries.
Building / Application (0/1)	Building: model building or data and ML infrastructure. Application: technical application, functional application, product and testing, governance and security. Reference: general tech.
PhD intensity (standardized)	Share of the title's U.S. postings that name a PhD or doctorate, standardized across titles weighting by U.S. postings; one standard deviation is 12.5 percentage points (Section 5.4).
Location cost (standardized)	Mean of the standardized title-level shares of U.S. postings requiring clearance or citizenship, on-site work, client contact and travel, minus the share advertised as remote (Appendix G).
Hiring level and timing	
Required experience (years)	Minimum years stated within forty characters of "experience"; a distinct job keeps its lowest stated requirement.
Junior / mid / senior (0/1)	0–2, 3–5 and 6 or more years, fixed before estimation; Westby–Modestino two-level split: fewer than four years.
Software developers (0/1)	O*NET 15-1252; comparison group: all other technical occupations.
Post (0/1)	On or after November 30, 2022 (monthly designs) or 2022Q4 (quarterly design).

Notes: KPMG, Deloitte and Infosys are excluded from every variable. Exact search terms for PhD intensity, location cost and experience are in Appendix D.

4.4 Descriptive statistics

Table 4 gives the five-country sample by country and category (Appendix Table 18 by year), and Appendix Figure 9 plots postings and distinct jobs by month. The sample is large in every country: in the median month it contains 8,689 U.S., 3,030 Indian, 968 British, 304 Mexican and 228 Philippine jobs, and no country has an empty month. Product and testing work makes up 16 percent of India's focal-category jobs and 9 percent of the United States'; model building 2.9 and 3.7 percent. India's share of the firms' combined U.S. and Indian focal-category jobs rose from 15.1 percent in 2019 to 35.0 percent in 2025; part of that rise is LinkUp's expanding coverage, which is why it is not an outcome.

Table 4: Distinct jobs by task category and country, 2019–2025

	Titles	United States	India	United Kingdom	Mexico	Philippines
Building						
LLM_BUILDER	1,035	27,366 (3.7)	6,593 (2.9)	2,314 (2.9)	428 (1.9)	444 (2.9)
DATA_MLOPS_ENGINEER	1,843	36,911 (5.0)	12,455 (5.5)	3,250 (4.0)	943 (4.2)	378 (2.4)
Application						
TECHNICAL_LLM_USER	1,895	90,294 (12.3)	20,634 (9.1)	5,661 (7.0)	1,405 (6.3)	685 (4.4)
FUNCTIONAL_LLM_USER	12,922	479,198 (65.4)	142,340 (63.1)	59,248 (73.3)	15,436 (69.5)	12,141 (78.7)
AI_UX_INTERACTION	1,461	63,550 (8.7)	36,227 (16.1)	6,635 (8.2)	3,503 (15.8)	1,129 (7.3)
AI_GOVERNANCE_RISK	1,588	34,854 (4.8)	7,360 (3.3)	3,774 (4.7)	494 (2.2)	659 (4.3)
Ordinary technical work						
GENERAL_TECH	3,116	54,033	11,164	3,599	1,766	1,796
AI categories, total	20,744	732,173	225,609	80,882	22,209	15,436
All raw postings		45,362,919	5,872,110	3,425,189	867,043	441,002

Notes: Distinct jobs (firm × normalized title × creation date) in titles of the fixed lookup (Section 5.2). In parentheses: percent of the country's focal-category jobs. "Titles" is the number of lookup titles in each category. "All raw postings" counts every unique LinkUp record of the sample firms in the country, in any title. KPMG, Deloitte and Infosys excluded.

Because every comparison is within a firm, it matters how hiring is spread across firms (Appendix Figure 10). No firm accounts for more than 2 percent of focal-category jobs, the ten largest for 13.8 percent and the hundred largest for 48.9 percent. What identifies the within-firm comparisons is the set of firms that hire in both the United States and a destination: 2,386 for India, 2,044 for the United Kingdom, 909 for Mexico and 551 for the Philippines.

Figure 1 shows the full-text sample. The two countries' stacks differ in shape: technical application is India's largest category (37 percent of its technical postings in 2025), while U.S. postings are spread more evenly and hold proportionally more model building and governance work. Building rose from 16.3 to 22.5 percent of U.S. technical postings between 2022 and 2025, and from 13.3 to 16.7 percent of Indian ones.

Appendix Tables 12 and 13 describe required experience. Indian postings ask for more experience than U.S. postings in almost every category and occupation: the mean is 4.0 to 4.6 years against 3.5 to 4.1, and the junior share (0–2 years) is 22 to 31 percent against 34

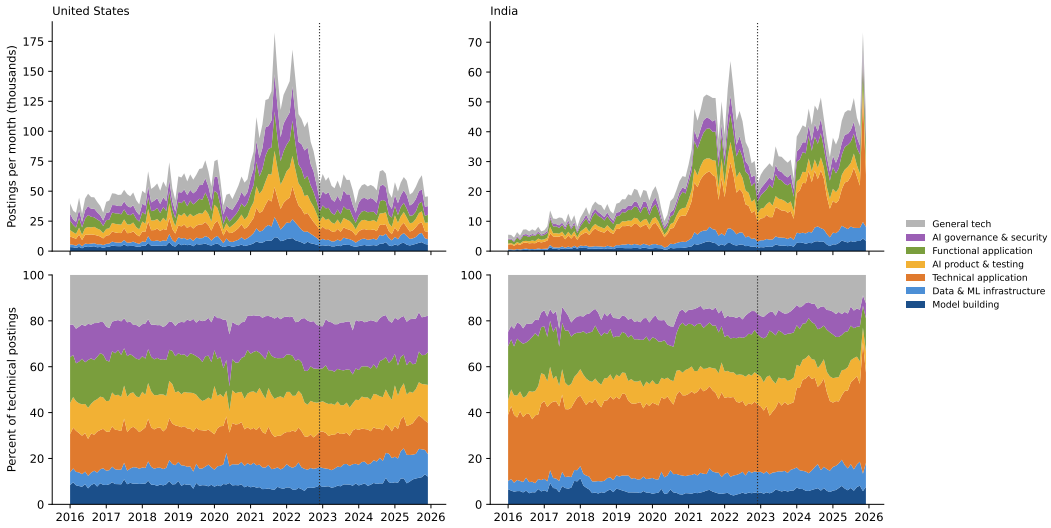


Figure 1: The technology stack in U.S. and Indian postings, 2016–2025. Top: technical postings per month (thousands), stacked by category of the stack. Bottom: each category's share of the country's technical postings that month. Categories are read from the full posting text (Section 5.1); Building (model building, data and ML infrastructure) is at the bottom of each stack. 11.1 million postings of the panel firms; KPMG, Deloitte and Infosys excluded. Counts mix hiring with LinkUp's scraping coverage, which grew over the period, especially in India; the last month (December 2025) shows a jump in Indian postings that may reflect a collection vintage. The dotted line marks the release of ChatGPT.

to 42. The gap is widest for software developers, where 21 percent of Indian postings are junior against 37 percent of U.S. ones. Ordinary technical work is the most junior category everywhere.

5 Measuring Tasks from Posting Text

Occupation codes cannot locate tasks (Section 2): they are too coarse, they cannot tell two jobs with the same title apart, and they are missing outside the United States and India. I therefore read the kind of technical work from the text of each posting, in two ways built for different comparisons. A full-text classifier, with category definitions that contain no AI vocabulary, labels every U.S. and Indian posting since 2016 and is the measure for everything that compares the two countries. A topic model, whose labels I fix at the level of job titles, follows the same work into the United Kingdom, Mexico and the Philippines and is the measure for the five-country results. This section describes both, the checks on them, and the task's skill intensity.

5.1 Seven categories read from the full text

For the United States and India, LinkUp supplies the full description of each posting. For each kind of technical work I write a one-sentence description without any AI vocabulary:

- *T1 Model building* (LLM_BUILDER): develops and trains predictive and machine-learning models, designs architectures, prepares training data, runs experiments, evaluates accuracy;
- *T2 Data and ML infrastructure* (DATA_MLOPS_ENGINEER): data pipelines and warehouses, feature stores, model deployment and serving, monitoring, distributed and cloud data infrastructure;
- *T3 Technical application* (TECHNICAL_LLM_USER): develops applications and services, connects systems through APIs, writes, tests and maintains code;
- *T4 Product and testing* (AI_UX_INTERACTION): product management, interface and interaction design, user research, customer-facing front ends;
- *T5 Functional application* (FUNCTIONAL_LLM_USER): enterprise and business systems, reporting and dashboards, business analysis and process improvement;
- *T6 Governance and security* (AI_GOVERNANCE_RISK): information-security controls, audits, risk frameworks, regulatory compliance and policy;
- *T7 General tech* (GENERAL_TECH): networks, servers, systems administration, cloud operations, help desk and desktop support.

The first two make up *Building*, the next four *Application*, and general tech is the benchmark. An analogy from manufacturing may help: Building is designing the engine and building the plant that makes it; Application is fitting the engine into cars, deciding where it is used, testing the finished car and certifying that it meets regulations; general tech keeps the factory's machines running. Because the descriptions name no AI tool or technique, the categories can be followed from 2016, before generative AI.

Each posting is split into consecutive windows of 350 words (at most eight), each window is embedded together with the title by a sentence-embedding model (BGE-large; Xiao et al. 2023), and the posting's vector is the mean of its windows. A posting is assigned to the category whose description is nearest after subtracting the corpus mean. Reading the whole posting matters: U.S. postings are about twice as long as Indian ones, and reading the full text rather than a fixed truncation changes about a fifth of the labels, equally in the two countries.

5.2 A topic model and a fixed title lookup for five countries

Outside the United States and India only titles are comparable, so for the five-country results I fix each title's category from its U.S. postings. The title's category comes from a topic model, described first.

The topic model, BERTopic (Grootendorst 2022), works in three steps. Only the third involves a judgment by the researcher.

First, each posting's title and the first 2,000 characters of its description are passed through a sentence-embedding model (all-MiniLM-L6-v2; Reimers and Gurevych 2019), a neural network trained on about a billion pairs of related sentences to place texts with similar meaning close together. For each posting it returns 384 numbers. No single coordinate means anything; what matters is distance. Two postings that describe adapting large language models end up close together even if one says "LLM fine-tuning" and the other "customizing foundation models," a match that no keyword list written in advance would make. An economist may think of the embedding as a vector of hedonic characteristics learned from how language is used.

Second, within each year, I partition the postings into fifty groups by k-means clustering on these coordinates. Fifty is deliberately much larger than the number of categories, so that work of different kinds is not forced into the same group. The model is fit on all 13.6 million U.S. and Indian technical postings with descriptions (9.96 million U.S., 3.62 million Indian), not only on the analysis sample.

Third, the groups are named. To read what a group is about, the model finds the words and two-word phrases that are *distinctive* to it, by class-based term-frequency–inverse-document-frequency weighting. I compare each group's twenty most distinctive terms with a vocabulary written in advance for each category and assign the group to the category with the most matches (Appendix D). Groups whose distinctive terms match no vocabulary are classified as ordinary technical work (general tech). Labelling whole groups rather than individual postings makes the classification depend on what the postings in a group have in common, not on any single word; an earlier keyword classifier in this line of work let generic business-intelligence vocabulary define "AI use" and inflated one category to two thirds of all AI-tagged postings.

Figure 2 shows the same structure in 2022 and 2025, and Figure 3 what the model sees in 2025. In the full 384-dimensional space, 70 percent of a posting's fifteen nearest neighbours belong to its own category, against about 17 percent if categories were assigned at random; the share ranges from 45 percent for functional application to 85 percent for testing and governance work.

A title is assigned to a category if it has at least 30 U.S. postings and at least 60 percent

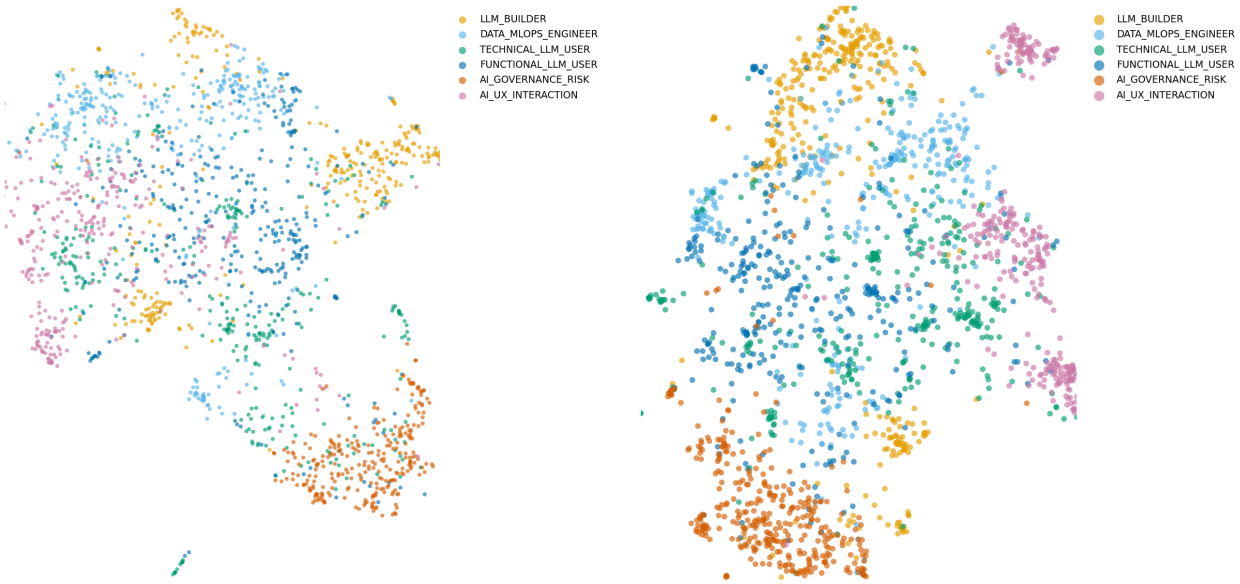


Figure 2: The topic model's task categories in 2022 (left) and 2025 (right). Each dot is a sampled U.S. or Indian technical posting, placed by a two-dimensional UMAP projection of its sentence embedding and coloured by the category of its cluster. The model is fit separately each year, so the axes are not comparable across panels.

Table 5: The six focal task categories of the title lookup

Category	Work performed	Largest titles
Building		
LLM_BUILDER	Make the models: train, tune and test them	network engineer; machine learning engineer
DATA_MLOPS_ENGINEER	Build the data pipelines and platforms models run on	cloud engineer; cloud architect
Application		
TECHNICAL_LLM_USER	Build software that uses models made by others	systems engineer; embedded software engineer
FUNCTIONAL_LLM_USER	Plan, deploy and manage AI-enabled systems in a business	project manager; program manager; business analyst
AI_UX_INTERACTION	Test AI-enabled products and design how users interact with them	test engineer; QA engineer
AI_GOVERNANCE_RISK	Keep AI-enabled systems secure and compliant	cyber security engineer; information security analyst
GENERAL_TECH	Technical work with no AI content (benchmark)	technical support specialist; desktop support technician

Notes: Largest titles by U.S. distinct jobs in the fixed title lookup (Section 5.2); Appendix Table 19 lists more. AI_UX_INTERACTION is dominated by testing and product engineering, and AI_GOVERNANCE_RISK by security governance. LLM_BUILDER includes network-engineering titles.

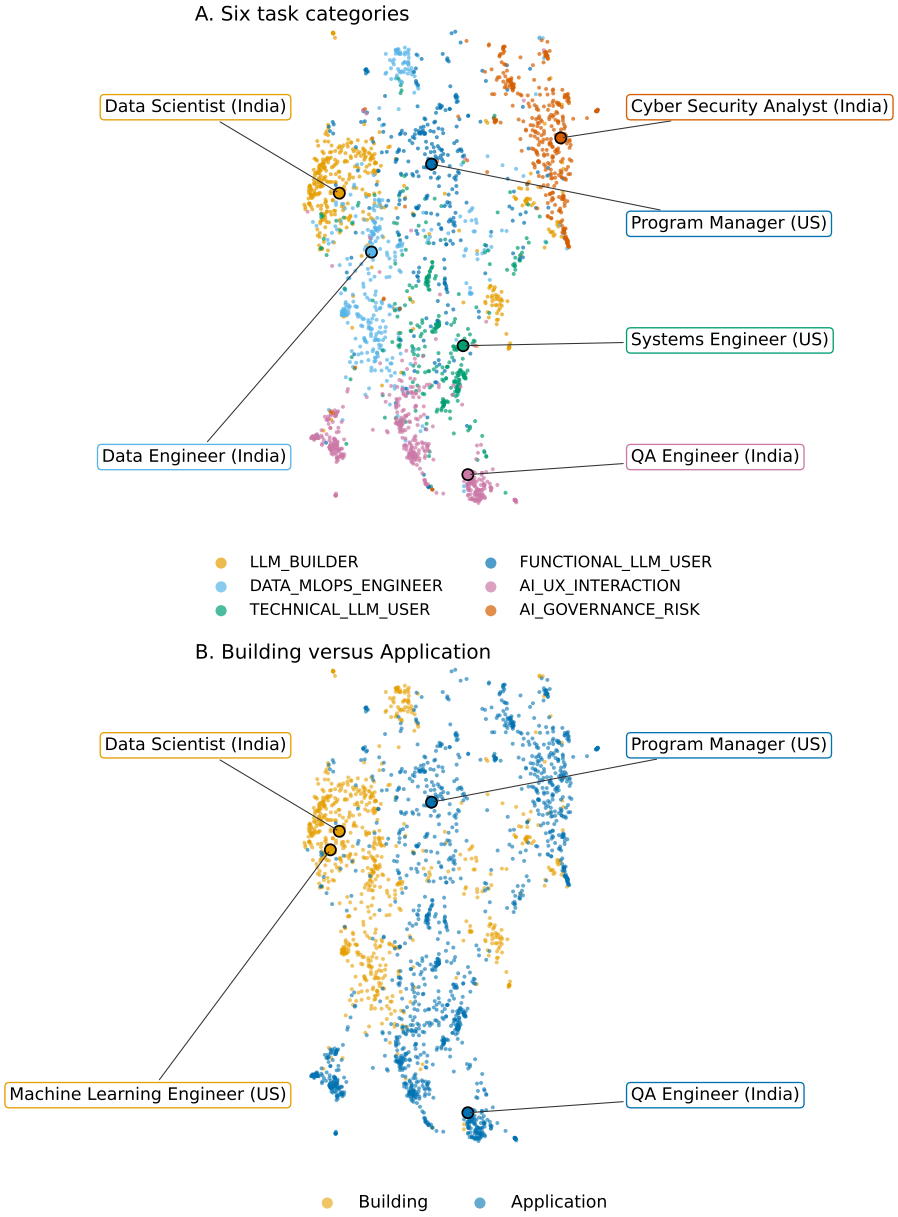


Figure 3: What the topic model sees: 2025 U.S. and Indian technical postings in embedding space. Each dot is a posting placed by a two-dimensional UMAP projection of its 384-dimensional MiniLM embedding (600 sampled per category). Panel A: colour by task category. Panel B: colour by Building versus Application. Outlined dots are labelled with the posting's actual job title and country. The axes have no units.

of them carry that category's label, pooling 2019–2025. The lookup contains 20,744 focal-category titles and 3,116 general-tech titles, and it labels a job identically in every year and country. The cost is coverage: it matches 3 to 8 percent of each country's postings (United States 4.1, India 7.5, United Kingdom 3.1, Mexico 3.1, Philippines 4.6), and generic titles such as "analyst" drop out. Two features matter for interpretation. Functional application is broad, 63 to 79 percent of focal-category jobs in every country, with project manager, program manager and business analyst as its largest titles. And the lookup's model-building category contains network engineering: titles containing "network" are 44.6 percent of its jobs (78 percent in the Philippines). The topic model's original labels (LLM_BUILDER and so on; Table 5) describe the groups' distinctive vocabulary, not the jobs; the lookup is best read as a classification of technical work by its position in the stack.

5.3 Checks on the categories

The Building–Application split follows the engineering account of production machine-learning systems, in which the code that trains a model is a small part of a deployed system and data collection, feature pipelines, serving and monitoring make up most of the effort (Sculley et al. 2015). A separate named-entity model (GLiNER) extracts skill mentions from posting text, which I classify against a reviewed lexicon of AI and machine-learning skills. Within Software Developers (O*NET 15-1252), model-building postings have the highest AI-skill share of their skill mentions in every year examined: 0.82 percent in 2019, 1.37 percent in 2022 and 2.06 percent in 2024, against at most 0.18, 0.45 and 0.94 percent in the other five categories. Within the lookup, no category except model building is dominated by one title: the largest title holds 4 to 9 percent of its category's jobs, and a small number of non-technical titles (2.5 percent of focal-category jobs) enter through firm-specific postings.

For the full-text categories, a check that needs no human coding uses titles whose category is not in doubt (machine-learning engineer, help-desk technician, SAP consultant and the like): 5,063 such postings in a test corpus of 24,558. With the title in the input, the labels match the expected category for 90.5 percent of them (macro average), with U.S. and Indian accuracy within 11 points in every category. With the title removed from the input and its phrase deleted from the body, accuracy falls to 73.9 percent, just under the 75 percent set in advance as a pass mark: the text alone identifies model building (80 percent), governance and security (88), general tech (90) and product and testing (82) well, but confuses technical with functional application. Appendix F compares the categories with job titles and O*NET occupations.

5.4 Skill intensity

The model's second task attribute, s_z, is how much of a task's labor must come from workers with graduate training. For every lookup title I compute, from 2.34 million U.S. postings with full text, the share that ask for a PhD or doctorate, and standardize it across titles. A PhD is named in 12.4 percent of U.S. model-building postings, 6.0 percent of product and testing, 5.0 percent of technical application and 3.7 percent of data and ML infrastructure postings, against 2.2 percent in general tech, 1.6 percent in functional application and 0.8 percent in governance and security. Skill intensity is unrelated to measured location cost (posting-weighted correlation across titles -0.01). The measure has a known flaw: some employers' templates list an education-substitution ladder ("Bachelor's and eight years, Master's and six, PhD and four"), so a mention of a PhD is not always a requirement.

6 Empirical Framework

6.1 Estimation strategy

The paper estimates four equations, all by Poisson pseudo-maximum likelihood (PPML) on counts. PPML is consistent if the conditional mean is correct, whether or not the counts are Poisson, and it keeps the many zero cells (Santos Silva and Tenreyro 2006). Coefficients carry subscripts that name their estimand, so that no symbol is reused for a different quantity.

The first equation describes where tasks go (P1–P4). For firm i, country $c \in \{\text{US}, \text{IN}, \text{UK}, \text{MX}, \text{PH}\}$, job title z and year t,

$$\mathbb{E}[V_{iczt}] = \exp\Big(\alpha_{izt} + \gamma_{ict} + \sum_{d} \mathbb{1}[c = d]\,\big(x_z'\beta_d^{\tau} + s_z\,\beta_d^{s}\big)\Big), \tag{9}$$

where V_{iczt} counts distinct jobs, x_z is the vector of indicators for the title's six focal categories (general tech is the reference), and s_z is standardized PhD intensity, with β_d^s pooled over 2019–2025 (year-by-year estimates are in Section 7.3). This is Equation (5): $\beta_{d,k}^{\tau}$ is destination d's price, in placement odds relative to the United States, of category k relative to general tech, and β_d^s its price of graduate intensity. The firm × title × year effect α_{izt} absorbs the firm's total demand for the task and every attribute of the title; the firm × country × year effect γ_{ict} absorbs the firm's scale in each country and with it LinkUp's coverage. Table 6 estimates the equation without the skill term, Table 7 with it. The hypotheses are $H_0\colon \beta_{d,k}^{\tau} = 0$ (P1), equality of the destinations' rankings of the $\beta_{d,k}^{\tau}$ (P2, assessed by Spearman correlations and, in Appendix G, by joint tests on measured attributes), $H_0\colon \beta_{d,B}^{\tau} - \beta_{d,A}^{\tau} = 0$ for the Building–Application gap (P3), and $H_0\colon \beta_d^s = 0$ together with

the predicted ordering of the β_d^s (P4).

The second follows India's position in the stack over time (P3, P5). On the U.S.–India full-text sample, with task categories k in place of titles,

$$\mathbb{E}[V_{ickt}] = \exp\big(\alpha_{ikt} + \gamma_{ict} + \mathbb{1}[c = \text{IN}]\,\beta_{kt}^{\text{IN}}\big), \tag{10}$$

where β_{kt}^{IN} is India's odds of receiving category k in year t, relative to general tech, for $t = 2016, \ldots, 2025$. P5 is the hypothesis that $\beta_{B,t}^{\text{IN}} - \beta_{A,t}^{\text{IN}}$ stays constant while β_{kt}^{IN} rises for every k.

The third asks whether generative AI moved tasks (P6). On firm × country × category × quarter counts of U.S. and Indian postings in 2018Q1–2025Q4 (1.10 million cells),

$$\mathbb{E}[V_{icku}] = \exp\big(\alpha_{iku} + \gamma_{icu} + \mathbb{1}[c = \text{IN}]\,(\beta_k^{\text{lev}} + \beta_k^{\text{trend}}\,u + \beta_k^{\text{break}}\,\text{Post}_u)\big), \tag{11}$$

for each category k in T1–T6, with general tech as the reference; u indexes quarters. β_k^{break} is the break at 2022Q4 in India's odds of receiving k, beyond the category's own trend. Under channel (ii) of Proposition 5, $H_0\colon \beta_k^{\text{break}} = 0$ holds for every k; under channel (i) every β_k^{break} equals $\theta\xi_{dt} > 0$. Relocation common to every task, support included, is absorbed by γ_{icu}.

The fourth measures entry-level hiring (P6, P6′). Following Westby and Modestino (2026), for firm f, country c, occupation group g (software developers or all other technical occupations), level l and month t, estimated by country,

$$\mathbb{E}[n_{fcglt}] = \exp\big(\alpha_{fcgl} + \varsigma_{fcgt} + \zeta_{clt} + \beta_J\,\text{Junior}_l \times \text{Post}_t \times \text{SW}_g + \beta_M\,\text{Mid}_l \times \text{Post}_t \times \text{SW}_g\big), \tag{12}$$

with senior (6+ years) as the reference level. α_{fcgl} absorbs each firm's level mix within each occupation group, ς_{fcgt} every shock to the firm's hiring in that group and month, and ζ_{clt} every shock common to a level in the country and month. β_J is the change after November 2022 in the log ratio of junior to senior software-developer postings, relative to the same ratio in the firm's other technical postings. The India − U.S. difference pools the arms with all fixed effects interacted with country. The main window is Westby and Modestino's, November 2021 to October 2023. The hypothesis is $H_0\colon \beta_J = 0$.

Standard errors are clustered by firm, the unit that decides where each job goes and that receives common shocks, and they are robust to heteroskedasticity. No sampling weights are used: the population is the firms' postings, not a sample of them. Two further tools guard against false positives. For Equation (11), inference compares the break at 2022Q4 with the same coefficient estimated at eleven placebo dates, 2019Q1 to 2021Q3, each on 2018Q1–2022Q3 so that every placebo has at least four quarters on either side; the rank

p-value is the share of the twelve dates whose break is at least as large in absolute value. For Equation (12), a family of ten designs was specified in advance, and I report Holm-adjusted p-values across the family together with the rule fixed beforehand: significant after adjustment in the U.S. arm with a null placebo. The specifications of Equations (10)–(12) were written down before estimation; the text flags the analyses that were not.

6.2 Identification strategy

The paper makes two kinds of claims, and they rest on different arguments.

The allocation estimates are descriptive equilibria, read through a model. Equations (9) and (10) do not estimate the effect of a shock. They measure how the same firm, in the same year, divides different tasks between the United States and a destination, and the model gives those comparisons an interpretation as prices of task characteristics. Their credibility rests on the fixed effects absorbing everything the model says is common to a firm's tasks in a country—wages, the general cost of operating there, the firm's own premium, LinkUp's coverage—so that what is left is the task-specific part. With four destinations they cannot separate distance, time zone, language and the wage gap, and I read the β_d^τ through each destination's profile (Section 2).

The generative-AI estimates use timing. Equations (11) and (12) ask whether something changed at ChatGPT's release, beyond each series' own trend and beyond what happened to comparison work in the same firms. The release date was set by OpenAI, not by the firms, and neither design uses a firm's own adoption of AI, which would be a choice. Their credibility rests on what else changed at the same time (Section 2). Shocks that changed a firm's overall hiring in a country—interest rates, layoffs, a general slowdown of Indian fresher hiring—are absorbed by γ_{icu} in Equation (11) and by ς_{fcgt} and ζ_{clt} in Equation (12). Shocks specific to a category or to junior software work are not, and the placebo dates, the pre-trend tests and the false release date are there to show whether such shocks were moving the series before the release. The placebo dates also measure power: they include the shift to remote work, which should have moved tasks if anything did. What the relocation test can detect is relocation that favors technical work over routine support, the ξ_{dt} of Proposition 5(i). Relocation that moved every kind of technical work abroad in the same proportion would change only κ_{dt}; it is absorbed by γ_{icu} together with LinkUp's coverage of India, and these data cannot detect it. The restriction matters less than it might seem, because the work generative AI reached first—writing and testing code—is far from routine support, but it bounds what the null result says, as does the possibility, which no placebo can exclude, of a category-specific shock of the opposite sign at the same date.

I take the three sources of statistical endogeneity in turn.

Reverse causality. For the allocation, the worry is that location shapes the measured task: an Indian posting may be written differently from a U.S. posting for the same work. The title lookup removes this for the five-country results, since a title's category is fixed from its U.S. postings alone, and so do the location-cost and skill-intensity measures, which use only U.S. text. For the full-text categories it remains possible, and I treat it as measurement error below. For the timing designs reverse causality is not a concern: the firms' hiring did not determine the release date.

Unobserved heterogeneity. The fixed effects absorb anything common to a firm's tasks within a country and period, and anything common to a task within a firm and period. What they cannot absorb is coverage that differs *by task* within a firm and country—if LinkUp scraped a firm's Indian engineering pages more completely than its Indian support pages, for example. Proposition 1 assumes this away; it is the one assumption of the allocation results that the data cannot check. In the levels design, the remaining threat is a shock specific to junior software work that coincided with the release; the pre-trend tests and the placebo date address it, and the comparison with non-software engineers in the pre-specified family gives a similar estimate, though not one that survives the multiple-testing adjustment.

Measurement error. The task labels are inferred. On postings whose titles leave the category in no doubt, the full-text labels are correct 90.5 percent of the time with the title in the input and 73.9 percent without it, and the title-blind accuracy gap between the countries reaches 19 to 20 points in three categories. Error that differs by country could manufacture or hide a U.S.–India difference. Its direction is known, however: correcting each country's category shares for its own confusion matrix widens India's Building shortfall from 6.4 to about 16 percentage points, so misclassification makes India's Building work look *more* like the United States', not less. Human coding of a stratified sample is under way (Appendix K). PhD intensity is measured with error because of substitution ladders; error common to all destinations attenuates every β_d^s by the same factor and leaves their ordering intact (Appendix I). Experience is parsed from text, and Indian employers ask for 0.87 more years for the same firm, title and year, so a fixed two-year cutoff picks a slightly higher rung in India; Section 7.2 redefines the rungs by each country's own distribution.

Spillovers (SUTVA). Three kinds of interference are possible. Within a firm and task, the allocation is a share, so a job placed in India is a job not placed in the United States; the model builds this in rather than assuming it away, and every coefficient is read as relative odds. Within a firm, AI that reaches one category could change hiring in another, which would contaminate the comparison categories; I test this by asking whether hiring in the never-exposed category moves with how much of the firm's work sat in the first-exposed

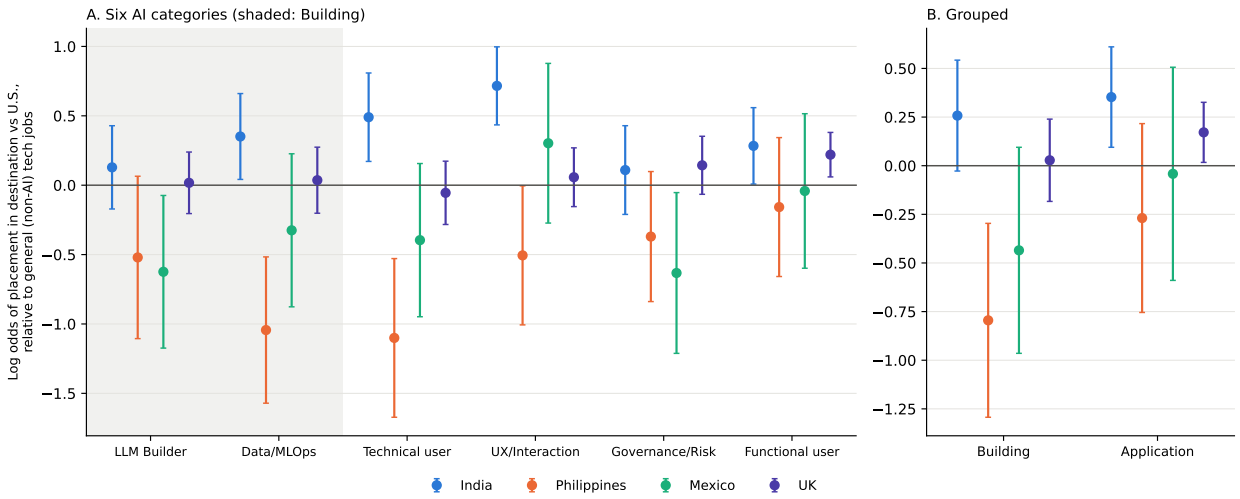


Figure 4: Placement of each focal task category in the destination rather than the United States, relative to ordinary non-AI technical work. Panel A: six categories (shaded: Building). Panel B: Building and Application grouped. 95 percent intervals; standard errors clustered by firm.

categories, and find no sign of it (+0.10, s.e. 0.24, in the U.S. arm), although the test is imprecise. If AI also reduced junior hiring in the comparison occupations, the triple difference understates the effect. Across firms, offshoring by one firm moves the common U.S. white-collar wage (Proposition 7); this is why the wage comparisons in Section 7.4 test whether group labor markets are integrated and do not measure the level of the productivity effect.

7 Results and Discussion

7.1 Main results

The results build up in four steps: which categories each destination receives (the simplest comparison), the same comparison with graduate intensity added, India's position over a decade, and finally the two tests of what generative AI did.

Which categories go where (P1–P3)

Table 6 and Figure 4 report the destination prices of the six focal categories, relative to general tech.

Firms offshore selectively (P1). India is the one destination that receives focal-category work in preference to ordinary technical work: product and testing by 0.72 log points ($t = 5.0$), technical application by 0.49 ($t = 3.0$), ML infrastructure by 0.35 ($t = 2.2$)

Table 6: PPML estimates of task placement by category and destination, relative to ordinary technical work

	India	Philippines	Mexico	United Kingdom
Building				
Model building	0.13	−0.52*	−0.62**	0.02
	(0.15)	(0.30)	(0.28)	(0.11)
Data and ML infrastructure	0.35**	−1.04***	−0.33	0.04
	(0.16)	(0.27)	(0.28)	(0.12)
Application				
Technical application	0.49***	−1.10***	−0.40	−0.05
	(0.16)	(0.29)	(0.28)	(0.12)
Product and testing	0.72***	−0.51**	0.30	0.06
	(0.14)	(0.26)	(0.29)	(0.11)
Governance and security	0.11	−0.37	−0.63**	0.14
	(0.16)	(0.24)	(0.30)	(0.11)
Functional application	0.28**	−0.16	−0.04	0.22***
	(0.14)	(0.26)	(0.28)	(0.08)
Grouped (separate regression)				
Building	0.26*	−0.79***	−0.43	0.03
	(0.15)	(0.25)	(0.27)	(0.11)
Application	0.35***	−0.27	−0.04	0.17**
	(0.13)	(0.25)	(0.28)	(0.08)
Building − Application	−0.10	−0.53***	−0.39***	−0.14*
	(0.10)	(0.15)	(0.11)	(0.08)

Notes: PPML of Equation (9) without the skill term, with destination × category interactions. Outcome: distinct jobs in firm × country × title × year cells, 2019–2025. Reference group: general tech (3,116 titles of non-AI technical work). 1,864,390 cells; fixed effects firm × title × year and firm × country × year. Standard errors clustered by firm in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and functional application by 0.28 ($t = 2.0$). The Philippines and Mexico show the reverse: their offshored technical work is predominantly ordinary support and IT work, and every focal category except, in Mexico, testing work is placed there less (the Philippines receives Building 0.79 log points less than general tech; Mexico receives product and testing 0.30 more, imprecisely). These are relative odds. In levels the United States remains the larger location for every category (Table 4).

Model building stays home where the destination offers only lower wages (P3). Building is placed abroad less than Application by 0.53 log points in the Philippines (s.e. 0.15) and 0.39 in Mexico (s.e. 0.11); the gap is smaller and marginal in the United Kingdom (0.14, s.e. 0.08). On lookup titles India offshores Building and Application at similar rates (-0.10, s.e. 0.10); read from the full text, it does not (below).

Destinations do not rank tasks alike (P2). India's top categories are product and testing and technical application; the United Kingdom's and the Philippines' are functional application and governance (the United Kingdom receives functional application 0.22 log points more than general tech, s.e. 0.08), and technical application is last in both. The Spearman correlation between India's and the United Kingdom's orderings of the six categories is -0.31, and between India's and the Philippines' -0.43; the Philippines and the United Kingdom order them almost identically (0.94), although one is a low-wage and the other a high-wage destination. The rankings track what each destination offers, not its wage, which a single GRH ranking cannot produce.

Skill intensity (P4)

Table 7 adds graduate intensity. A one-standard-deviation increase in PhD intensity lowers the log odds that a job is placed in the destination rather than the United States in all four destinations: by 0.10 in India (s.e. 0.02) and the United Kingdom (0.03), 0.24 in Mexico (0.05) and 0.35 in the Philippines (0.06). Adding location cost changes neither set of coefficients, and the gradient holds within task categories (row 3), so it does not restate the category results.

The ordering prediction holds in part. The coefficients are negative in the three low-wage destinations, and of the six pairwise orderings the model implies, four hold: the Philippines, where doctorates are rarest, penalizes PhD intensity most, and the United Kingdom penalizes it less than the Philippines (difference 0.25, s.e. 0.07) and Mexico (0.14, s.e. 0.06). Both failures involve India, whose coefficient is as close to zero as the United Kingdom's (difference 0.006, s.e. 0.04) although doctorates are rarer there than in Mexico; India's far larger supply of master's graduates may be the margin that matters. And the United Kingdom's coefficient is negative, where its slightly larger doctoral supply predicts a positive one. One reading

Table 7: PPML estimates of task placement by skill intensity and destination

	India	Philippines	Mexico	United Kingdom
A. PhD intensity, linear (per standard deviation)				
(1) PhD intensity	-0.099^{***}	-0.352^{***}	-0.241^{***}	-0.105^{***}
	(0.022)	(0.063)	(0.046)	(0.032)
(2) With location cost (Appendix G)	-0.119^{***}	-0.344^{***}	-0.243^{***}	-0.104^{***}
	(0.023)	(0.062)	(0.046)	(0.032)
Location cost	-0.262^{***}	-0.150^{*}	-0.049	0.030
	(0.050)	(0.083)	(0.071)	(0.048)
(3) Within category	-0.108^{***}	-0.257^{***}	-0.203^{***}	-0.089^{***}
	(0.020)	(0.055)	(0.045)	(0.034)
B. Share of the title's U.S. postings naming a PhD (reference: none)				
0–5 percent	0.257^{***}	0.143	0.319^{***}	0.374^{***}
	(0.078)	(0.107)	(0.095)	(0.051)
5–15 percent	0.115	-0.110	-0.331^{**}	0.186^{***}
	(0.099)	(0.171)	(0.131)	(0.061)
15–30 percent	-0.014	-0.411	-0.440^{**}	-0.099
	(0.145)	(0.258)	(0.181)	(0.087)
Over 30 percent	-0.352^{***}	-1.645^{***}	-0.905^{***}	-0.364^{**}
	(0.123)	(0.317)	(0.242)	(0.162)
C. Building − Application, relative to ordinary technical work				
Without PhD intensity	-0.095	-0.526^{***}	-0.393^{***}	-0.143^{*}
	(0.098)	(0.148)	(0.112)	(0.080)
With PhD intensity	-0.042	-0.434^{***}	-0.294^{***}	-0.091
	(0.094)	(0.146)	(0.110)	(0.082)

Notes: PPML of Equation (9) with destination × skill-intensity interactions; firm × title × year and firm × country × year fixed effects; standard errors clustered by firm in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. PhD intensity: share of the title's U.S. postings naming a PhD, standardized across titles (one standard deviation is 12.5 percentage points). Rows (1), (3) and panels B–C use all six focal categories and ordinary technical titles (1,080,940 cells after separation); row (2) uses the AI titles, for which location cost is defined (1,013,952 cells). Row (3) adds destination × category effects. Panel B: one regression; bins fixed before estimation; 71.7, 20.9, 3.4, 1.5 and 2.5 percent of U.S. postings are in the none, 0–5, 5–15, 15–30 and over-30 percent bins. Panel C: destination × Building and destination × Application, without and with destination × PhD intensity.

is a location cost of graduate-intensive work common to all destinations—closeness to U.S. research teams, data and compute—which shifts every β_d^s down by the same amount. Panel C confirms P3: controlling for PhD intensity shrinks the Building–Application gap by about a fifth in the Philippines (−0.53 to −0.43) and a quarter in Mexico (−0.39 to −0.29); the rest is a Building-specific location cost.

The stack in India, 2016–2025 (P3, P5)

Figure 5 estimates Equation (10) on 391,132 cells from 4,677 firms. India rose through the whole stack: relative to general tech, its odds of receiving every other category were higher in 2025 than in 2016—grouped, from −0.28 (s.e. 0.17) to +0.21 (0.07) for Building and from +0.14 (0.13) to +0.63 (0.19) for Application. Technical application is the category India receives most (+0.54 in 2016, +1.39 in 2025), and governance and security is the least offshored but converging (−0.66 to −0.11). The gap at the top did not close: Building trailed Application by 0.42 log points in 2016 (s.e. 0.08) and by 0.42 in 2025 (0.17), after narrowing to about 0.22–0.25 in 2022–23. The flat gap is the pattern of Proposition 4(iii): a fall in location costs common to every category above support lifts all of them and leaves the gap where it was. India's role has not moved toward model building in particular; it moved up every category at once. A rise of every category against general tech is also what one would see if India's odds of receiving support work fell, if support were automated or brought home; the estimates cannot separate the two, but either reading leaves the Building–Application gap, which does not involve general tech, unchanged.

Did generative AI move tasks? (P6)

Table 8 answers the paper's question. No category shows a break at the release, relative to general tech, that is unusual against the placebo dates. The point estimates are positive but no larger than the typical placebo break; for technical application, the category code generation reached first, the break is +0.10 (s.e. 0.13) with a rank p of 0.83. The one borderline category is governance and security (rank p 0.08), whose convergence began in 2023 (Figure 5); it is the category code generation reaches least, so neither channel of Proposition 5 predicts a break there in particular. At the release the data are consistent with channel (ii), location-neutral labor saving, and not with channel (i), relocation.

The test can detect relocation of the kind it is built to find. The largest breaks in the series are not at 2022Q4 but at the placebo dates of 2020Q2–2021Q2, when the shift to remote work lowered the cost of performing technical work away from the office: +0.2 to +0.7 log points across categories, largest in 2020Q4 for technical application (+0.68, s.e.

Table 8: PPML estimates of breaks in India's odds of receiving each category at the release of ChatGPT, 2022Q4

Category	Break at 2022Q4	Conv. p	Mean |placebo|	Rank p
Building				
Model building	0.125	0.30	0.28	0.83
	(0.122)			
Data and ML infrastructure	0.172	0.34	0.26	0.92
	(0.179)			
Application				
Technical application	0.096	0.47	0.26	0.83
	(0.134)			
Product and testing	0.239*	0.07	0.16	0.33
	(0.133)			
Functional application	0.252	0.12	0.28	0.67
	(0.164)			
Governance and security	0.269*	0.05	0.10	0.08
	(0.138)			

Notes: β_k^{break} of Equation (11). PPML on firm × country × category × quarter counts of U.S. and Indian technical postings, 2018Q1–2025Q4, categories from the full posting text; fixed effects firm × category × quarter and firm × country × quarter; regressors India × category, India × category × trend and India × category × Post (2022Q4 on), reference category general tech. Standard errors clustered by firm in parentheses. Placebo breaks at each quarter 2019Q1–2021Q3, each estimated on 2018Q1–2022Q3 (11 dates); rank p: share of the 12 dates with an absolute break at least as large as the true one. Pre-specified, with one deviation recorded before the estimates were seen: firm × country × category effects did not converge and were replaced by the common India × category terms. Stars use the conventional p: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

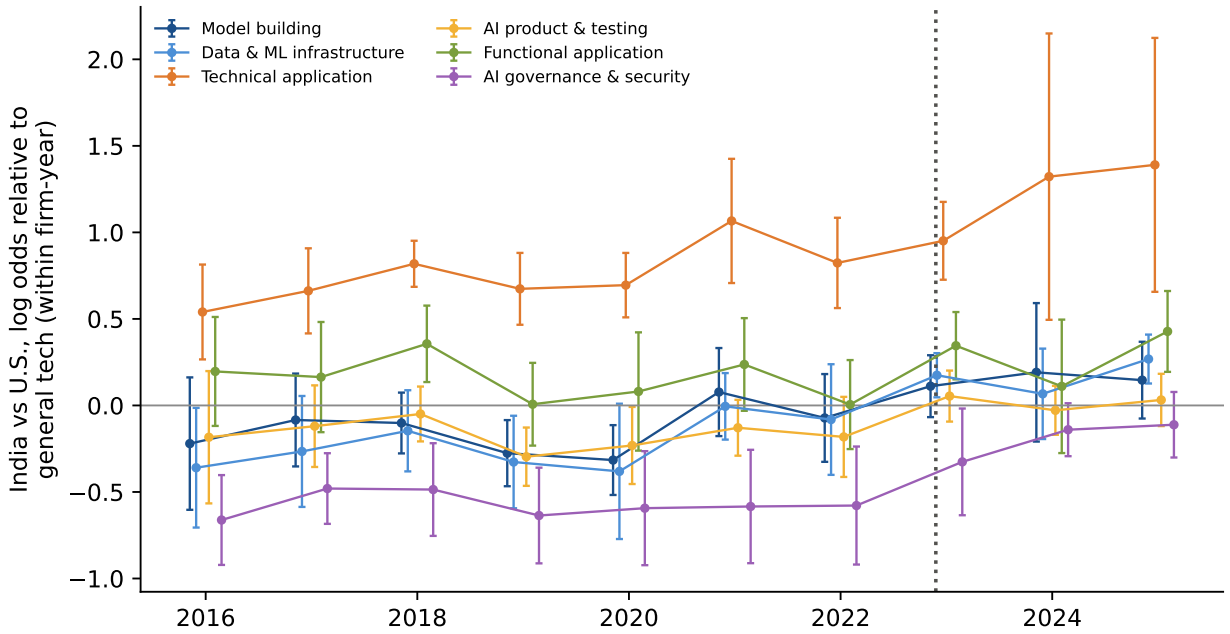


Figure 5: India's odds of receiving each category rather than the United States, relative to general tech, by year, 2016–2025. PPML coefficients β_{kt}^{IN} of Equation (10) with firm × category × year and firm × country × year fixed effects; 95 percent intervals, standard errors clustered by firm. Categories labelled from the full posting text (Section 5.1); 11.1 million U.S. and Indian technical postings, 391,132 cells, 4,677 firms; KPMG, Deloitte and Infosys excluded. The dotted line marks the release of ChatGPT.

0.20) and model building (+0.49, s.e. 0.16). In the model this is channel (i) at work. This finding was not pre-specified, and a proper test of it needs its own design; what it shows here is that breaks of this size can be detected in these data, and none occurred in 2022–23. Against that benchmark the null is informative: in every category the 95 percent interval of the break at the release excludes +0.68, the largest remote-work break, and in model building and technical application it excludes breaks above +0.37.

Entry-level hiring at home and abroad (P6, P6′)

Table 9 reports Equation (12). In the U.S. arm, junior software-developer postings fell by 0.215 log points (s.e. 0.063), 19 percent, relative to senior postings and to junior postings in other technical occupations; the Holm-adjusted p-value is 0.002. Mid-level postings did not move (−0.043, s.e. 0.060), so the decline is concentrated on the rung open to workers with at most two years of experience. With Westby and Modestino's two-level split the estimate is −0.155 (s.e. 0.052), against their −0.136 (s.e. 0.058) on nearly all U.S. vacancies: in the window both papers use, the two designs agree. In the same firms' Indian operations there is no such decline: +0.006 (s.e. 0.054) on their definition of junior and −0.021 (s.e. 0.073) on

mine. The 95 percent interval for the first, $[-0.10, 0.11]$, excludes a decline as large as the U.S. one; the interval for the second, $[-0.16, 0.12]$, just reaches it. The difference between the arms is +0.194 (s.e. 0.087; Holm-adjusted $p = 0.052$).

Table 9: PPML estimates of junior and mid-level software-developer postings after ChatGPT, U.S. and Indian operations of the same firms

	United States	India	India − U.S.
A. Westby–Modestino window, November 2021 – October 2023			
Junior (0–2 years) × Post × SW	−0.215***	−0.021	0.194**
	(0.063)	(0.073)	(0.087)
Mid (3–5 years) × Post × SW	−0.043	0.067	0.109
	(0.060)	(0.053)	(0.074)
Holm-adjusted p, junior	0.002	0.779	0.052
B. Placebo: false date November 2021, window November 2020 – October 2022			
Junior × Post × SW	0.010	0.009	−0.001
	(0.078)	(0.077)	(0.127)
C. Pre-trends, 2019Q1–2022Q2 (joint Wald test, p-value)			
Junior × SW	0.28	0.12	
Mid × SW	0.06	< 0.001	
D. Two-level split (junior < 4 years), same window			
Junior × Post × SW	−0.155***	0.006	0.160***
	(0.052)	(0.054)	(0.061)
Firms (panel A)	3,190	2,155	3,452

Notes: β_J and β_M of Equation (12). PPML on firm × occupation group × level × month counts of technical postings with a stated minimum experience (LinkUp; KPMG, Deloitte and Infosys excluded). SW: software developers (O*NET 15-1252); comparison: all other technical occupations. Senior (6+ years) is the reference level. Fixed effects firm × group × level, firm × group × month and level × month (by country); India − U.S. pools the arms with all fixed effects interacted with country. Standard errors clustered by firm in parentheses; stars use unadjusted p-values (Holm-adjusted p reported separately): * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel C: quarterly event study, reference 2022Q3. Panel D: S1 of the pre-specified family (Appendix Table 15).

The pooled estimate summarizes a short episode (Figure 6). Before 2022Q4 the junior coefficients are jointly indistinguishable from zero in both arms ($p = 0.28$ in the United States, $p = 0.12$ in India; the U.S. mid-level pre-trend test is marginal, $p = 0.06$), and a false release date a year earlier gives estimates of zero (Panel B). After the release the U.S. junior coefficient is close to zero for two quarters, falls to −0.24 (s.e. 0.11) in 2023Q3 and −0.12 in 2023Q4, and returns to around zero from 2024. The Westby–Modestino window compares the year after the release with a year that includes unusually strong junior hiring in late 2021 (+0.24 in 2021Q4), which accounts for part of the −0.215. A dip followed by recovery

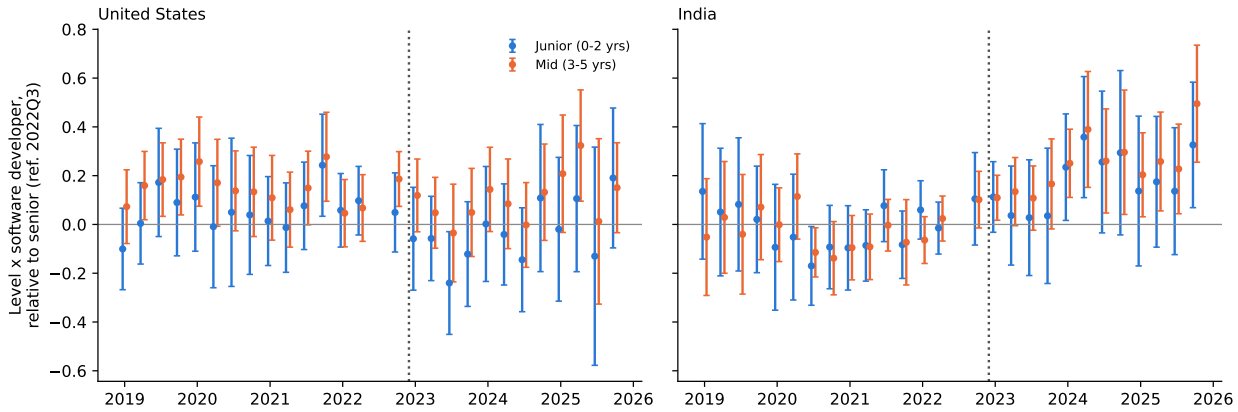


Figure 6: Junior (0–2 years) and mid-level (3–5 years) software-developer postings relative to senior postings and to other technical occupations, by quarter, U.S. and Indian operations of the same firms (reference 2022Q3). Coefficients from the event-study version of Equation (12), 2019–2025, with 95 percent intervals; standard errors clustered by firm. The dotted line marks the release of ChatGPT.

is the response Friebel et al. (2026) predict when AI raises productivity: a short-run freeze in junior hiring, after which the firm restores its pyramid because the juniors it hires today are the seniors it needs later.

What the model gets right and wrong

Table 10 collects the model's predictions, the test of each and the verdict. Seven hold (one only on wages paid), one holds in part, one is consistent with the evidence but rests on a design whose pre-trend test fails, the cross-firm test of the mechanism behind the U.S. dip fails, and one proposition is not tested. The verdicts are uneven in a way that is informative: the allocation predictions, which follow from the structure of location costs, hold; the predictions that depend on measured graduate supply hold only in part; and the model says more about where the junior effect should appear than the data can confirm about why it appeared.

7.2 Robustness checks

Adding the measured location-cost requirements of Appendix G to Equation (9) leaves the category coefficients essentially unchanged: location cost explains which titles within a category go abroad, and the category captures what the text measure does not. The single ranking is rejected on the measured requirements as well (Appendix G.3). For India the two task measures disagree about the Building gap because the lookup's model-building category contains network engineering (Section 5.2), which India receives readily. On lookup titles

Table 10: The model's predictions, how each is tested, and what the data show

Prediction	Test	Result	Verdict
P1. Selective offshoring	$\beta^{\tau}_{d,k} \neq 0$ in Eq. (9)	India receives four focal categories more than general tech (+0.28 to +0.72); the Philippines and Mexico receive most of them less (Table 6)	Holds
P2. Destination-specific rankings	Rankings of $\beta^{\tau}_{d,k}$ across d; joint tests on measured attributes	Spearman correlation of India's ranking with the UK's −0.31, with the Philippines' −0.43; single ranking rejected on measured attributes (Appendix G.3)	Holds
P3. Model building stays home where graduates are scarce	$\beta^{\tau}_{d,B} - \beta^{\tau}_{d,A}$, with and without $s_z \beta^{s}_{d}$	Gap −0.53 (Philippines), −0.39 (Mexico), −0.14 (UK); about 0.4 for India on the full text; skill explains a fifth to a quarter (Table 7, Panel C)	Holds
P4. Sorting on skill, ordered by graduate supply	$\beta^{s}_{d} < 0$ and $\beta^{s}_{UK} > \beta^{s}_{MX} > \beta^{s}_{IN} \gtrsim \beta^{s}_{PH}$	Negative in all four; four of six pairwise orderings hold. Both failures involve India ($\approx$ UK, predicted below Mexico); UK < 0 where its doctoral supply predicts > 0 (Table 7)	Partly
P5. A destination climbs the stack without closing the gap at the top	β^{IN}_{kt} rises for all k; $\beta^{\mathrm{IN}}_{B,t} - \beta^{\mathrm{IN}}_{A,t}$ constant (Eq. (10))	Every category rises 2016–2025; the gap is 0.42 in 2016 and 0.42 in 2025 (Figure 5)	Holds
P6. Generative AI saves labor without moving tasks	$\beta^{\mathrm{break}}_{k} = 0$ (Eq. (11)); $\beta_J < 0$ at home only (Eq. (12))	No break, relative to support, unusual against placebos (rank p 0.08–0.92); every interval excludes the largest remote-work break. U.S. junior −0.215 (0.063), India −0.021 (0.073) (Tables 8, 9). Relocation common to all work is not identified	Holds
P6′. Temporary dip at home; shift to the middle offshore	Event study; rung shares	U.S. dip in 2023, gone by 2024; Indian mid share 46 to 65 percent, but its pre-trend test fails (Figures 6, 7)	Consistent, descriptive
Mechanism: offshoring selection explains which firms' U.S. juniors fell	Junior × Post × SW × firm's India share	−0.086 (0.083), wrong sign; continuous dose null (Section 7.4)	Fails
P7. U.S. pay in offshored tasks keeps pace with white-collar pay (one labor market), not behind it (segmented markets)	Pay growth vs. white-collar, 2019–2025	Building +25.4, Application +24.9, white-collar +24.4 percent (Figure 8)	Holds
P7′. Building and Application pay rise alike	Same	Within half a point on wages paid; posted pay in Building grew more slowly (9.3 percent), about half of it explained by starting pay	Holds on wages paid
Migration steepens the skill gradient (Prop. 3)	—	Migration is not observed by firm	Not tested

Notes: Predictions from Section 3.8. Standard errors in parentheses, clustered by firm. P4's ordering uses the graduate supplies of Section 2. "Consistent, descriptive" marks evidence that matches the prediction but does not pass the design's own pre-trend test.

India's Building–Application difference was −0.38 (s.e. 0.17) in 2019 and closed by 2023 (−0.002); read from the duties, the gap is about 0.4 log points and the 2022–23 narrowing was temporary. The misclassification correction points the same way, so I treat the full-text estimate as the answer for India.

The penalty on graduate-intensive work survives the known flaws of the measure. Dropping titles whose PhD share is 0.9 or more, the clearest template cases, gives linear coefficients of −0.07, −0.33, −0.25 and −0.09 for India, the Philippines, Mexico and the United Kingdom; dropping the top decile of titles by clearance requirements gives −0.09, −0.37, −0.23 and −0.08; keeping titles with at least 100 U.S. postings gives −0.11, −0.48, −0.28 and −0.13. Master's-degree intensity has a weaker gradient of the same sign (−0.05 to −0.32), and when the PhD, master's and research measures enter together, PhD intensity carries the effect.

The junior result does not depend on where the experience cutoff falls. Because Indian employers ask for 0.87 more years for the same firm, title and year (39,089 pairs, s.e. 0.21), a fixed two-year threshold picks a slightly higher rung in India. Defining the rungs by each country's own pre-2022 terciles, which puts the junior cutoff at three years in India and two in the United States, leaves the results unchanged: −0.214 (s.e. 0.063) in the U.S. arm, +0.026 (s.e. 0.059) in India, +0.240 (s.e. 0.075) for the difference. I added this check after the pre-specified designs.

Other estimators and windows bound the size of the U.S. effect. Averaged over 2022Q4–2023Q3, the U.S. effect is −0.08 with a 95 percent interval of −0.23 to 0.08, and Rambachan–Roth bounds that allow post-release violations of parallel trends as large as the largest pre-period one widen it further. A synthetic difference-in-differences on the junior share of firm × occupation units with postings in every quarter of 2020–2024 (120 software-developer and 357 other technical units in the United States) gives −0.005 (s.e. 0.011); those are the largest, most regular posters, so it speaks to a narrower set of firms. Over the longer horizon to 2025 the U.S. junior estimate is −0.095 (s.e. 0.084). These checks bound the U.S. effect: it is a 2023 dip, not a durable fall.

Westby and Modestino (2026) report that the market for junior software developers had not recovered by early 2025, while the U.S. junior coefficient here returns to about zero from 2024. The two papers agree about the decline in the window both estimate; they differ about its persistence, and three differences in design can account for that. First, their evidence on persistence is the raw ratio of senior to junior software vacancies, which has no comparison group and so also carries the 2023–24 contraction of technology hiring; the event study here nets out junior postings in other technical occupations. Second, their regressions use month and metropolitan-area fixed effects, so a change in which firms post moves their

estimate; every comparison here is within firm. The pooled shares of Figure 7 show why this matters: they contain a 2023 dip in Indian software that the within-firm estimate does not. Third, their data cover nearly all U.S. employers, and they find the effect significant only in the largest firm-size quintile; this paper covers large multinationals with Indian operations, which may have restored their junior pipelines sooner than other employers. They also impute experience for the 36 percent of software vacancies that do not state it, which I do not. Re-estimating their specification on these data, adding the differences one at a time, would show which one matters; I have not yet done so.

Appendix Table 15 reports all ten pre-specified designs. The software-developer indicator is the only member with the expected sign that meets the rule fixed in advance. Using non-software engineers as the comparison group gives −0.147 (s.e. 0.072), and occupations whose tasks people more often hand to Claude, and more often for automation, show larger junior declines (−0.053 and −0.057 per standard deviation of the Anthropic Economic Index measures, s.e. 0.025); neither survives the Holm adjustment. The GPT-4 exposure ratings show no gradient. With the full-text category in place of the occupation group, U.S. junior postings fell in model building (−0.25, s.e. 0.14) and technical application (−0.15, s.e. 0.09), but no category survives the Holm adjustment across the six (adjusted $p \geq 0.42$). A stacked event study that dates each category by the month its core task first became doable by a language model is too imprecise to confirm or contradict the occupation result (U.S. junior coefficient −0.01, s.e. 0.17).

7.3 Heterogeneity

The category prices differ by destination in the way the destinations' profiles suggest, and the formal version on measured requirements makes this precise (Appendix Table 22): India avoids client-facing, on-site and clearance-requiring work; the United Kingdom receives *more* client-facing and on-site work and less work requiring U.S. clearance; Mexico, near and in U.S. time zones, receives 0.12 log points more on-site work than India ($t = 3.0$). Each destination receives the tasks whose location costs it is best placed to absorb.

Along the skill distribution, the skill penalty is a doctoral margin, not a gradient through the whole range of tasks (Table 7, Panel B). Relative to titles whose U.S. postings never name a PhD, mostly business and support roles, titles that name one occasionally are placed abroad *more* often in every destination; these are engineering and analytical roles. The penalty appears at the top: titles in which more than 30 percent of U.S. postings name a PhD—applied scientists, machine-learning engineers, biostatisticians—are placed abroad 30 percent less in India and the United Kingdom, 60 percent less in Mexico and 81 percent less

in the Philippines. A version on the posting-weighted rank of the PhD share, which weights the many low-share titles as much as the few high-share ones, changes sign (India +0.07, s.e. 0.03; United Kingdom +0.12, s.e. 0.02), for the reason Panel B shows. Over time, India's PhD-intensity coefficient rose from −0.24 (s.e. 0.06) in 2019 to −0.06 in 2022 and has stayed between −0.07 and −0.12 since; a general fall in offshoring costs cannot produce this, since it is absorbed by γ_{ict}.

Across countries, the two arms of the levels design differ beyond the junior coefficient, and the difference depends on the comparison. Figure 7 plots the share of postings at each rung by year (Appendix Table 14 gives the numbers). Between 2022 and 2025 the junior share of Indian software-developer postings fell by 10.3 percentage points and that of other Indian technical postings by 7.2, against 3.7 and 2.0 in the United States. The postings went to the middle rung, whose share rose by 18.4 points in Indian software and 10.2 in other Indian technical work, while Indian software also lost seniors (−8.2). In the United States the middle barely moved. The shift is the diamond of Proposition 8, offshore and not at home.

The triple difference does not see this loss because it measures juniors against *seniors*, in software against other occupations. In the pooled shares, the change in the log ratio of junior to senior postings between 2022 and 2025 is −0.09 in Indian software and −0.16 in other Indian technical work: because Indian software lost seniors as well, its juniors held up slightly better against seniors than juniors elsewhere. The United States shows the opposite pattern in 2023, when the software ratio fell by 0.32 and the ratio in other occupations by 0.03, a gap that had closed by 2025. The pooled Indian shares also contain a 2023 dip in software juniors (−0.22 against other occupations) that the within-firm estimate does not (−0.021, s.e. 0.073), so that dip reflects which firms posted, or which postings stated experience, rather than a change within firms.

Table 11 sets the estimated comparisons side by side. Against seniors and other occupations, Indian software juniors did not fall. Within the Indian operations, junior postings fell relative to senior ones in Application work (−0.176, 95 percent interval [−0.30, −0.06]), not in model building (+0.036), and imprecisely in general tech (−0.325, s.e. 0.226). And firm by firm, the junior share of AI jobs in India did not fall by more than that of the same firm's non-AI technical and engineering jobs (+2.1 percentage points, s.e. 7.3, 109 firms). The Indian operations therefore lost junior postings relative to the United States and to the middle rung, in application and support work, across every technical occupation; they did not lose them in software relative to seniors, or in AI jobs relative to the same firm's other jobs.

Three cautions apply to the Indian shift toward the middle: it comes in 2023–24, not

Table 11: Junior postings after November 2022 in the U.S. and Indian operations of the same firms, by comparison

Outcome	Compared with	United States	India	India, 95% interval
Junior (<4 yrs), software vs other technical	seniors; other technical occupations	−0.155	0.006	[−0.101, 0.112]
		(0.052)	(0.054)	
Junior (0–2 yrs), software vs other technical	seniors; other technical occupations	−0.215	−0.021	[−0.164, 0.123]
		(0.063)	(0.073)	
Junior (<4 yrs), Application titles	seniors in the same group	−0.004	−0.176	[−0.295, −0.058]
		(0.046)	(0.061)	
Junior (<4 yrs), Building titles	seniors in the same group	0.101	0.036	[−0.188, 0.260]
		(0.105)	(0.114)	
Junior (<4 yrs), general tech titles	seniors in the same group	−0.120	−0.325	[−0.767, 0.118]
		(0.145)	(0.226)	
Junior share (<4 yrs, pp), Application jobs	same firm's non-AI technical jobs	1.2	2.1	[−12.2, 16.5]
		(2.3)	(7.3)	

Notes: Each row is a separate comparison of junior postings before and after November 2022; standard errors clustered by firm in parentheses (heteroskedasticity-robust in the last row). Rows 1–2: PPML triple difference of Equation (12), Westby–Modestino window; row 1 is S1 of the pre-specified family (Appendix Table 15), row 2 is Table 9, Panel A. Rows 3–5: PPML difference in differences of junior against senior postings within each task group of the title lookup, by country. Row 6: firm-weighted change in the junior share of the firm's Application jobs minus the same change for its non-AI technical and engineering jobs, in percentage points (Appendix Table 17). KPMG, Deloitte and Infosys excluded. Source: notebook `india_junior_comparisons`.

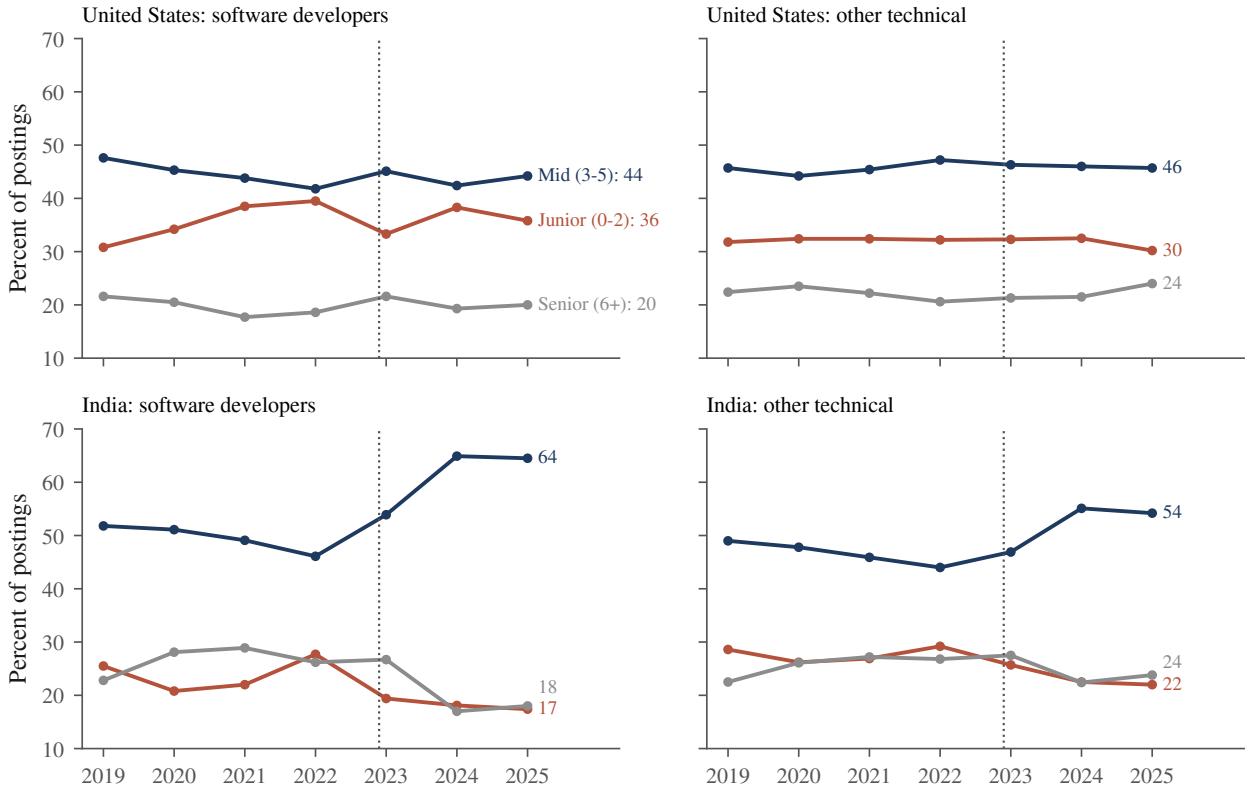


Figure 7: Share of technical postings at each rung of the experience ladder, by year, U.S. and Indian operations of the same firms. Junior: 0–2 years of required experience; mid: 3–5; senior: 6 or more. Software developers: O*NET 15-1252; other technical: all other technical occupations. Postings with an O*NET code and a stated minimum experience requirement, pooled across firms (6.0 million postings; KPMG, Deloitte and Infosys excluded). The dotted line marks the release of ChatGPT. Source: notebook `india_junior_comparisons`.

at the release; India's mid-level coefficients trended up before 2022 ($p < 0.001$ for the joint pre-trend test); and it coincides with a slowdown in fresher hiring across Indian technical work. I read it as a change in the structure of the firms' Indian operations that the model predicts, not as an estimated effect of ChatGPT. The data do not separate AI saving junior labor in codified work from a general contraction of Indian fresher hiring.

By type of firm, the twenty firms whose product is AI—model developers, compute providers and AI software firms—advertise model building outside the United States less than AI application, in India as well as the United Kingdom, unlike the average multinational (Appendix J). The British estimate survives the exclusion of any single firm; the Indian one is suggestive only.

7.4 Mechanisms

Why the U.S. dip and not an Indian one

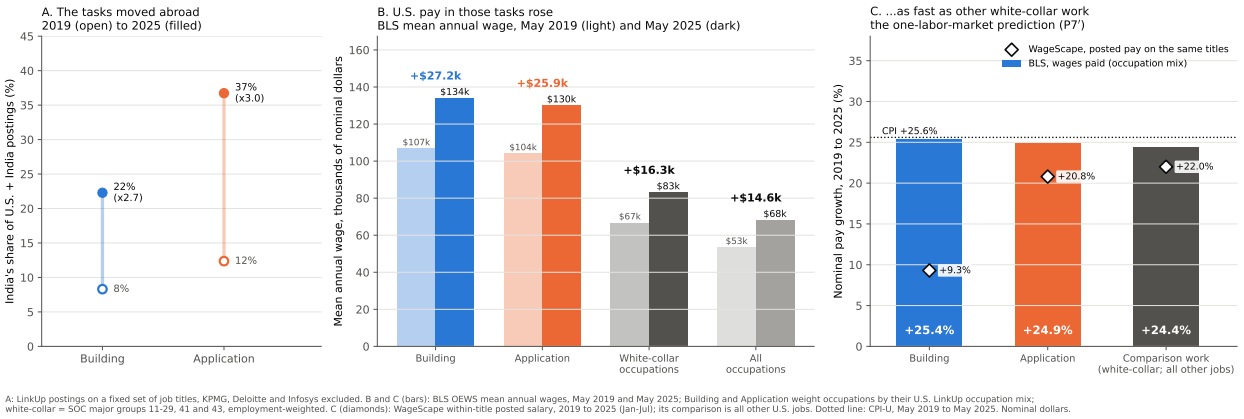
The model offers one mechanism for the country contrast. Junior hiring in a country falls by the task-weighted saving of junior labor, $\Delta \log J_c \approx -\sum_z \nu_{zc}\psi_z$ (Equation (6)), and the weights come from the allocation of Section 7.1. U.S. occupations keep a mix of codified tasks and tasks tied to the United States or to doctoral training, so exposure differs across U.S. occupations and software development stands out. Offshoring selection sends the codified slice of every occupation to India, so exposure is high and similar across Indian occupations, and the software-versus-other contrast has little to detect. The task-level evidence fits: in India the post-2022 junior decline is concentrated in application work and absent in model building, and it is as large in non-AI technical jobs as in AI-related ones.

I tested this mechanism directly, in a design specified in advance, and it fails. If offshoring selection explains the U.S. dip, a firm that had already sent more codified work to India should have less of it left at home and a smaller U.S. junior decline. Exposure is the firm's India share of its U.S.-plus-India Application postings in January 2019–October 2022, standardized across 990 firms, and it enters Equation (12) through Junior × Post × SW × exposure. The U.S. junior interaction is −0.086 (s.e. 0.083), the wrong sign, and in the top exposure tercile the U.S. declines are larger, not smaller. An exploratory event study shows that the top tercile's junior difference is a pre-existing trend (pre-trend $p = 0.03$), while its mid-level difference has flat pre-trends ($p = 0.73$) and averages −0.33 (s.e. 0.14) over the first year after the release—a lead for further work, not a finding. A second pre-specified design replaces the post-release indicator with a continuous dose, the log training compute of the best publicly usable language model in each month times each category's substitution and complementarity shares from the Anthropic Economic Index. The two shares are correlated at 0.9994 across the sixteen category × country cells, so they are not separately identified; their identified sum is −0.015 in the U.S. arm and +0.021 in India. The design shows no relation between AI capability, weighted by exposure, and junior hiring in the more exposed categories. Two instruments for the firm's own offshoring and for posting-level AI use are weak (first-stage $F \leq 3.3$; Appendix Table 16), so the paper does not decompose the U.S. effect into offshoring and automation. What the mechanism explains is the contrast between the two arms; it does not explain the differences among firms.

Offshoring and U.S. pay (P7, P7′)

In the model, offshoring affects U.S. pay only through the productivity effect, and with one white-collar labor market pay in offshored tasks keeps pace with other white-collar pay

(Proposition 7). The alternative is GRH's labor-supply effect: if task groups are segmented labor markets, workers released from offshored tasks lower their own group's wage, so pay in those tasks falls behind comparable work, and further where India's share rose more. The comparison therefore tests whether group labor markets are integrated. It does not measure the productivity effect, and because the home share is itself endogenous to U.S. wages, a co-movement of pay and offshoring would not identify it either. I measure wages *paid* from the BLS Occupational Employment and Wage Statistics, weighting each occupation's mean wage by its share in each task group's U.S. postings, and wages *advertised* from WageScape on the same lookup titles. All pay figures are nominal.



A: LinkUp postings on a fixed set of job titles, KPMG, Deloitte and Infosys excluded. B and C (bars): BLS OEWS mean annual wages, May 2019 and May 2025; Building and Application weight occupations by their U.S. LinkUp occupation mix; white-collar = SOC major groups 11-29, 41 and 43, employment-weighted. C (diamonds): WageScape within-title posted salary, 2019 to 2025 (Jan-Jul); its comparison is all other U.S. jobs. Dotted line: CPI-U, May 2019 to May 2025. Nominal dollars.

Figure 8: Offshoring and U.S. pay in the offshored tasks, 2019–2025. Panel A: India's share of U.S.-plus-India LinkUp postings in each task group, on a fixed set of job titles. Panel B: BLS OEWS mean annual wage, May 2019 and May 2025, nominal; Building and Application weight occupations by their U.S. LinkUp occupation mix; white-collar is SOC major groups 11–29, 41 and 43, employment-weighted. Panel C: the same growth in percent (bars), with WageScape posted salaries on the same titles (diamonds), whose comparison is all other U.S. jobs; dotted line: CPI-U, May 2019 to May 2025. KPMG, Deloitte and Infosys excluded.

Figure 8 sets the two margins side by side. On a fixed set of titles, India's share of these firms' U.S.-plus-India postings rose from 12.4 to 36.7 percent in Application and from 8.3 to 22.3 percent in Building between 2019 and 2025, a threefold rise in each. Over the same six years the mean annual wage rose from \$106,800 to \$133,900 in Building and from \$104,100 to \$130,000 in Application: 25.4 and 24.9 percent, against 24.4 percent for white-collar occupations and 25.6 percent for consumer prices. Both parts of P7′ hold. The two groups rose alike, within half a point, although India's share rose by 24 points in one and 14 in the other, and both rose with other white-collar work rather than behind it. The comparison is with white-collar work, not all workers, because tight post-pandemic labor markets compressed wages from below (Autor et al. 2023): median wages rose 33 to 45 percent in food service, personal care, transportation and healthcare support (Appendix

Figure 12).

The pattern repeats inside the technology industries and the ten largest software metros, on BrightQuery cuts of the same OEWS data for May 2019–May 2023 (Appendix Figure 14): the two groups rose 18 to 21 percent, 1 to 4 points behind white-collar occupations in 2023, a gap that had closed by 2025 nationally. On posted pay for the same Application titles (Appendix Figure 11), pay rose 20.8 percent against 22.0 percent for all other U.S. jobs (-1.0 log points, s.e. 1.4). At the level of individual titles posted pay grew slightly more slowly where India's share rose more (-0.042 log points per unit of India share, s.e. 0.014): a ten-point rise in India's share corresponds to about 0.3–0.4 percent lower relative posted pay, small next to the 20 to 25 percent rise in pay itself. Posted pay in Building grew more slowly, 9.3 percent; about half of that gap reflects starting pay, since better-paid titles received smaller raises throughout the U.S. market in this period (Appendix Figure 13), and OEWS shows no difference between the groups.

No cut shows the pattern that segmented markets with a dominant labor-supply effect predict. Under one white-collar labor market the evidence should not show a net relative *gain* for these workers either, since the shared gain is differenced out of every comparison with other white-collar work; its level is not identified here. Nor is the wage wedge ω_{id} of the model, since no source here measures Indian wages.

8 Conclusion

Picture the technical work of a large American firm as a stack, with model building at the top and help desks at the bottom. These firms split the stack across countries. They keep at home the work tied to the United States and the work that needs doctoral training, and they send abroad application, testing and infrastructure work that needs neither. Each destination takes a different slice: India the engineering side of application and infrastructure, the United Kingdom and the Philippines the business-facing side and governance. Over the past decade India climbed the whole stack at once, and its odds of receiving model building trailed its odds of receiving application work by the same 0.42 log points in 2025 as in 2016. When generative AI arrived, the split did not move: measured against routine support, India's share of each kind of work shows no break at ChatGPT's release. What moved, for about a year, was hiring at the bottom of the U.S. software ladder. The facts fit a technology that saves labor in codified work, which changes who is hired at each rung, not where the rungs are.

For policy, the results caution against reading offshoring as a one-for-one loss of U.S. jobs. The work that leaves is selected, the United States remains the main location of every

category, and U.S. pay in the work that moved fastest kept pace with other white-collar pay. A back-of-the-envelope calculation with Proposition 6 gives the size of what is at stake. The U.S. share of Application postings fell from 87.6 to 63.3 percent between 2019 and 2025, a fall of 0.325 in logs, and the U.S. share of Building postings from 91.7 to 77.7 percent, a fall of 0.166. By Proposition 6, the productivity effect of that much offshoring, at fixed output prices, is $0.325/\theta$ in Application and $0.166/\theta$ in Building. The placement elasticity θ is not identified here: at $\theta = 4$ the effect is about 8 and 4 log points of pay, and at Eaton and Kortum (2002)'s preferred value for goods trade, 8.28, about 4 and 2. Because part of each fall in the home share reflects LinkUp's expanding coverage of India, these are upper bounds. In the model, at fixed output prices, a restriction on offshoring would forgo that gain and would bind on tasks whose relocation did not lower U.S. pay. These data show no group of U.S. workers whose relative pay fell by more than a fraction of a percent as their tasks moved, so they identify no clear winner from such a restriction; the losers would include the U.S. white-collar workers who share the productivity effect. Restricting high-skilled immigration would push graduate-intensive work abroad, since for that work hiring the migrant and offshoring the task are substitutes (Glennon 2024): the foreign share of new graduate entrants into U.S. AI jobs fell from 70 to 54 percent when student visas were curtailed in 2020–21 (Pandit and Nambiar 2026). And the first entry-level effect of generative AI fell on the U.S. bottom rung of software development, the training ground for the senior engineers whose work these firms keep at home. Early-career training policies act on exactly that margin, and, unlike restrictions on where firms place work, they do not ask firms to forgo the productivity effect.

Three pieces of evidence would sharpen these conclusions, and each is within reach. The task categories are inferred from text; human coding of them, now under way, will show whether the measure errs differently in the two countries. The evidence is on postings; linking postings to the workers hired and to their pay would measure the productivity effect at the level of the firm and show whether the dip in junior postings was a dip in hires. And the model's account of why the dip appeared at home has not yet been confirmed across firms; firm-level variation in the cost of offshoring or of hiring migrants, such as changes in visa rules, would separate automation from relocation in the entry-level effect.

References

Acemoglu, D., Autor, D., Hazell, J., and Restrepo, P. (2022). Artificial intelligence and jobs: Evidence from online vacancies. *Journal of Labor Economics*, 40(S1):S293–S340.

Antràs, P., Garicano, L., and Rossi-Hansberg, E. (2006). Offshoring in a knowledge economy. *Quarterly Journal of Economics*, 121(1):31–77.

Arkolakis, C., Costinot, A., and Rodríguez-Clare, A. (2012). New trade models, same old gains? *American Economic Review*, 102(1):94–130.

Autor, D., Dube, A., and McGrew, A. (2023). The unexpected compression: Competition at work in the low wage labor market. NBER Working Paper 31010, National Bureau of Economic Research.

Blinder, A. S. and Krueger, A. B. (2013). Alternative measures of offshorability: A survey approach. *Journal of Labor Economics*, 31(2, Part 2):S97–S128.

Brynjolfsson, E., Chandar, B., and Chen, R. (2025). Canaries in the coal mine? Six facts about the recent employment effects of artificial intelligence. Working paper, Stanford Digital Economy Lab, November 2025.

Chandar, B. and Klein Teeselink, B. (2026). How does AI change labor demand? Evidence from 41 countries. Stanford University and King's College London, September 2026.

Costinot, A. (2009). An elementary theory of comparative advantage. *Econometrica*, 77(4):1165–1192.

Costinot, A. and Vogel, J. (2010). Matching and inequality in the world economy. *Journal of Political Economy*, 118(4):747–786.

Dekle, R., Eaton, J., and Kortum, S. (2007). Unbalanced trade. *American Economic Review*, 97(2):351–355.

Deming, D. and Kahn, L. B. (2018). Skill requirements across firms and labor markets: Evidence from job postings for professionals. *Journal of Labor Economics*, 36(S1):S337–S369.

Dingel, J. I. and Neiman, B. (2020). How many jobs can be done at home? NBER Working Paper 26948, National Bureau of Economic Research.

Eaton, J. and Kortum, S. (2002). Technology, geography, and trade. *Econometrica*, 70(5):1741–1779.

Eloundou, T., Manning, S., Mishkin, P., and Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702):1306–1308.

Friebel, G., Huang, Y., Li, J., Shukla, S., and Zhang, A. (2026). Pyramids, diamonds, and oscillations: AI and the structure of internal labor markets. Working paper, April 10, 2026.

Garicano, L. and Rossi-Hansberg, E. (2006). Organization and inequality in a knowledge economy. *Quarterly Journal of Economics*, 121(4):1383–1435.

Glennon, B. (2024). How do restrictions on high-skilled immigration affect offshoring? Evidence from the H-1B program. *Management Science*, 70(2):907–930.

Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*.

Grossman, G. M. and Rossi-Hansberg, E. (2008). Trading tasks: A simple theory of offshoring. *American Economic Review*, 98(5):1978–1997.

Hershbein, B. and Kahn, L. B. (2018). Do recessions accelerate routine-biased technological change? evidence from vacancy postings. *American Economic Review*, 108(7):1737–1772.

Hosseini, S. M. and Lichtinger, G. (2025). Generative AI as seniority-biased technological change: Evidence from U.S. résumé and job posting data. Working paper, SSRN 5425555.

Liu, R. and Trefler, D. (2011). A sorted tale of globalization: White collar jobs and the rise of service offshoring. NBER Working Paper 17559, National Bureau of Economic Research.

Pandit, H. and Nambiar, M. (2026). Who builds AI? the global supply of AI workers, migration, gender, and the firms that employ them. Working paper, Northeastern University and Boston College.

Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992.

Santos Silva, J. M. C. and Tenreyro, S. (2006). The log of gravity. *Review of Economics and Statistics*, 88(4):641–658.

Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., and Dennison, D. (2015). Hidden technical debt in machine learning systems. In *Advances in Neural Information Processing Systems 28 (NIPS 2015)*, pages 2503–2511.

Westby, S. and Modestino, A. S. (2026). The impact of generative AI on job opportunities for junior software developers. Manuscript submitted to Management Science, MS-0001-1922.65.

Xiao, S., Liu, Z., Zhang, P., and Muennighoff, N. (2023). C-Pack: Packaged resources to advance general Chinese embedding. arXiv:2309.07597.

A Additional descriptive statistics



Figure 9: Postings and distinct technical jobs per month, by country, log scale. Panel A: every unique LinkUp record of the sample firms. Panel B: distinct jobs (firm × normalized title × creation date) in the six focal categories and ordinary technical work. The dotted line marks the release of ChatGPT (November 30, 2022). Levels partly reflect LinkUp's scraping coverage and are not used as outcomes.

Table 12: Required experience by category and country, distinct jobs, 2019–2025

Layer	Country	Jobs stating (%)	Mean years	Median	Junior (%)	Senior (%)
Model building	United States	62	4.1	3	34.3	20.3
	India	64	4.3	3	26.7	20.5
	United Kingdom	22	3.6	3	33.3	11.4
	Mexico	52	3.9	3	29.6	12.6
	Philippines	48	3.3	3	37.4	8.4
ML infrastructure	United States	67	4.1	3	36.3	21.4
	India	67	4.6	4	22.1	25.2
	United Kingdom	23	4.3	3	28.4	18.9
	Mexico	53	4.1	4	22.9	15.1
	Philippines	46	3.3	3	45.7	10.9
Technical application	United States	64	4.1	3	34.7	23.0
	India	69	4.5	4	24.6	27.4
	United Kingdom	18	4.1	4	27.0	14.9
	Mexico	57	3.3	3	34.2	9.4
	Philippines	53	3.1	3	45.0	8.8
Functional application	United States	52	3.8	3	37.3	16.7
	India	54	4.2	4	31.1	22.7
	United Kingdom	18	4.0	3	32.6	15.3
	Mexico	42	3.5	3	37.7	11.1
	Philippines	49	3.4	3	43.7	9.9
AI testing and UX	United States	66	3.5	3	41.8	14.7
	India	65	4.0	3	27.5	19.1
	United Kingdom	18	4.1	3	30.3	13.7
	Mexico	47	3.5	3	32.8	13.5
	Philippines	53	3.4	3	40.8	10.6
AI governance and security	United States	67	4.0	3	34.2	20.4
	India	62	4.1	3	31.1	19.0
	United Kingdom	19	4.3	3	27.1	15.2
	Mexico	46	3.5	3	35.1	9.3
	Philippines	55	3.0	3	42.4	5.8
Ordinary technical work	United States	55	2.8	2	58.4	7.3
	India	65	3.1	2	52.1	12.5
	United Kingdom	20	2.9	2	50.1	5.0
	Mexico	41	2.1	2	70.2	3.8
	Philippines	52	2.2	2	73.7	1.8

Notes: Distinct jobs (firm × normalized title × creation date) in the fixed title lookup. "Jobs stating": share with a parsed minimum experience requirement; the remaining columns use those jobs. Junior: 0–2 years; senior: 6 or more. KPMG, Deloitte and Infosys excluded.

Figure 9 plots the sample by month, Figure 10 shows how focal-category jobs are spread across firms, and Tables 12 and 13 describe required experience by category and by occupation.

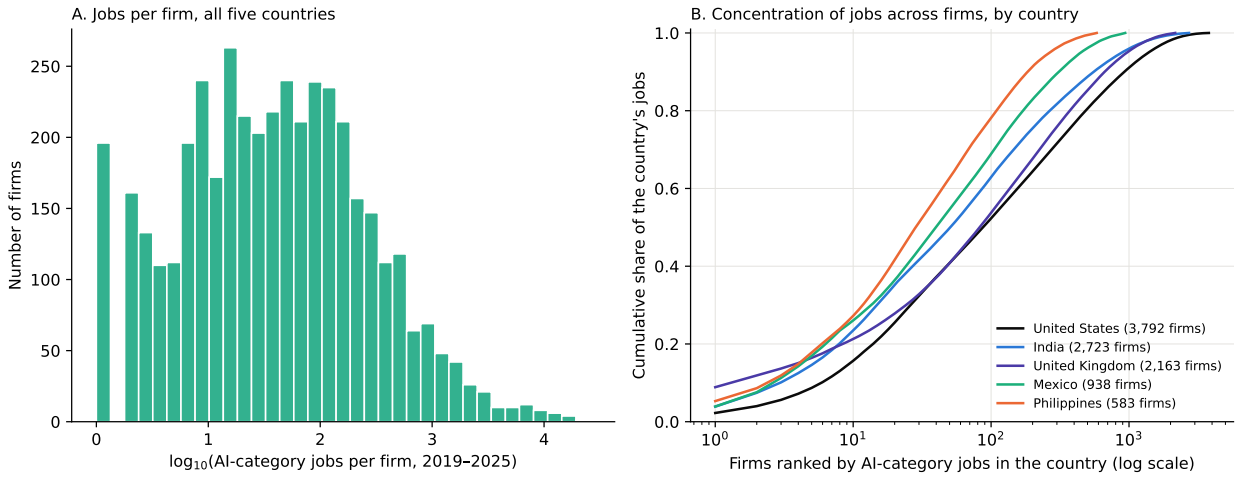


Figure 10: Distribution of focal-category jobs across firms. Panel A: distinct focal-category jobs per firm, 2019–2025, all five countries, log scale (4,210 firms). Panel B: cumulative share of each country's focal-category jobs held by its largest firms, firms ranked by jobs in that country.

Table 13: Required experience by occupation, United States and India, technical postings 2019–2025

	United States			India		
Occupation (O*NET)	Mean years	Junior (%)	Senior (%)	Mean years	Junior (%)	Senior (%)
Software Developers	3.9	36.6	19.5	4.4	21.0	23.2
Management Analysts	4.0	34.8	19.8	4.2	31.3	22.8
Computer and Information Systems Managers	5.5	15.7	39.6	5.7	11.0	31.4
Information Security Analysts	4.3	31.2	22.1	4.5	22.1	24.2
Computer User Support Specialists	2.9	52.7	8.8	3.4	40.4	13.0
Market Research Analysts and Marketing Specialists	3.5	40.9	12.4	3.4	38.4	13.4
Data Scientists	3.9	37.2	17.3	4.1	30.5	20.4
Information Technology Project Managers	4.6	23.4	25.5	5.7	13.6	42.9
Computer Network Architects	5.2	20.5	32.5	6.0	14.7	42.5
Computer Systems Analysts	3.7	33.5	14.2	4.4	23.7	26.7
Network and Computer Systems Administrators	4.3	29.6	22.8	4.5	20.2	25.4
Software Quality Assurance Analysts and Testers	3.7	37.9	14.9	4.1	24.6	19.7

Notes: Technical postings with an O*NET code and a parsed minimum experience requirement; the twelve largest occupations by U.S. postings. Junior: 0–2 years; senior: 6 or more.

Table 14 gives the shares plotted in Figure 7 for selected years.

Table 14: The shape of the ladder: shares of software-developer and other technical postings at each rung, U.S. and Indian operations

		Software developers			Other technical occupations		
	Year	Junior	Mid	Senior	Junior	Mid	Senior
United States	2019	30.8	47.6	21.6	31.8	45.7	22.4
	2022	39.5	41.8	18.6	32.2	47.2	20.6
	2023	33.3	45.1	21.6	32.3	46.3	21.3
	2024	38.3	42.4	19.3	32.5	46.0	21.5
	2025	35.8	44.2	20.0	30.2	45.7	24.0
India	2019	25.5	51.8	22.8	28.6	49.0	22.5
	2022	27.7	46.1	26.2	29.2	44.0	26.8
	2023	19.4	53.9	26.7	25.7	46.9	27.5
	2024	18.1	64.9	17.0	22.5	55.1	22.4
	2025	17.4	64.5	18.0	22.0	54.2	23.8

Notes: Percent of technical postings with an O*NET code and a stated minimum experience requirement, by year (LinkUp; KPMG, Deloitte and Infosys excluded). Junior: 0–2 years; mid: 3–5; senior: 6 or more. Software developers: O*NET 15-1252. Rows sum to 100 within each occupation group.

B The levels design: the pre-specified family and the instruments tried

Table 15 reports the ten designs specified in a pre-analysis plan written before estimation (available from the author), for the U.S. arm, estimated on the same cells (firm × country × occupation × level × month, two levels split at four years) and fixed effects, with Holm-adjusted p-values across the ten. The pre-specified rule required a Holm-adjusted $p < 0.05$ and a placebo with $p \geq 0.10$; it did not state a sign. S3 meets it with a positive coefficient: human raters place software developers near the bottom of technical exposure, so S3 restates S1 on an inverted scale. The Anthropic Economic Index measures (S5–S7) use task statements coded in O*NET-SOC 2010, mapped to 2019 codes with the official crosswalk. No member is significant in the Indian arm.

Table 15: Pre-specified family of exposure designs, U.S. arm, Westby–Modestino window

Design	Estimate	Holm p	Placebo	Meets rule
S1 Software developers vs other technical	-0.155^{***}	0.025	0.030	Yes
	(0.052)		(0.053)	
S2 GPT-4 exposure (β), per s.d.	-0.035	0.538	-0.019	No
	(0.026)		(0.031)	
S3 Human exposure (β), per s.d.	0.067^{***}	< 0.001	-0.005	Yes (wrong sign)
	(0.016)		(0.023)	
S4 GPT-4 direct exposure (α), per s.d.	-0.044^{*}	0.347	-0.016	No
	(0.026)		(0.028)	
S5 AEI usage intensity, per s.d.	-0.053^{**}	0.227	-0.010	No
	(0.025)		(0.029)	
S6 AEI automation share, per s.d.	-0.057^{**}	0.145	-0.004	No
	(0.025)		(0.023)	
S7 AEI automation, with usage	-0.048	0.538	0.029	No
	(0.036)		(0.051)	
S8 Software developers vs engineers	-0.147^{**}	0.227	0.031	No
	(0.072)		(0.071)	
S9 Software developers vs hardware/network	-0.249^{***}	0.026	0.208^{**}	No (placebo)
	(0.084)		(0.106)	
S10 Firm-level GPT-4 exposure	-0.013	0.538	0.009	No
	(0.010)		(0.008)	

Notes: PPML, Junior (< 4 years) $\times$ Post $\times$ exposure, fixed effects firm $\times$ occupation $\times$ level, firm $\times$ occupation $\times$ month and level $\times$ month; S10 is a firm-level weighted regression of the change in the junior share on firm exposure. Placebo: false date November 2021. Standard errors clustered by firm in parentheses; stars use unadjusted p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16 summarizes two instruments, each specified before estimation and reported whatever it showed. The first instruments the change in India's share of the firm's technical postings, with the change in the U.S. junior share as the outcome (one observation per firm); the second instruments whether a posting names a generative-AI tool. Neither has a first-stage F statistic above 3.3.

Table 16: Instruments tried for the levels design

Instrument	Endogenous variable	First-stage F	Units
Shift-share: pre-period occupation shares $\times$ leave-one-out change in India's share of LinkUp postings by occupation	Change in India share of the firm's postings	0.7	1,062 firms
Public-code abundance (The Stack) of the programming languages a posting names $\times$ Post	Posting names a generative-AI tool	3.0 (U.S.), 3.3 (India)	2.9 million postings

Notes: Robust (firm-level designs) or firm-clustered (posting-level design) first-stage F statistics. The language instrument's reduced form fails its placebo and pre-trend test. Each instrument was specified in a pre-analysis plan before estimation.

C Junior hiring in AI and non-AI technical work

India's information technology and services sector reduced fresher and junior hiring in 2023 and 2024. Such a contraction would lower junior hiring across all technical work in India, AI-related or not, from about the time of ChatGPT's release. To separate an AI-specific change from a general one, I compare AI jobs (Application and Building) with non-AI technical and engineering jobs at the same firms: GENERAL_TECH titles plus mechanical, civil, electrical, chemical, structural, manufacturing, industrial, process, aerospace, automotive, HVAC, piping and controls engineers. The comparison is made within firm, so that any shock to a firm's junior hiring in a country is differenced out, and it is estimated both with the two-level split of Westby and Modestino (2026) and with three levels (junior 0–2 years, mid 3–5, senior 6 or more), fixed before estimation.

Because a PPML specification with firm × group × level, firm × group × period and firm × level × period fixed effects leaves each firm's twelve cells nearly saturated and does not converge, I estimate the triple difference directly on shares. For each firm with at least five jobs in both the AI group and the comparison group, before and after November 2022, I compute the change in the share of its AI jobs at a given level minus the same change for its comparison jobs, and average across firms weighted by AI jobs, with heteroskedasticity-robust standard errors. Firm-specific shocks to junior hiring cancel within firm.

Table 17: Change in the junior share of AI jobs relative to non-AI technical and engineering jobs, same firm

	United States	India	Mexico
Two levels (junior < 4 years)			
Application, junior	1.15	2.14	5.56
	(2.29)	(7.32)	(3.47)
Building, junior	−2.71	8.03	—
	(4.25)	(8.36)	
Three levels (junior 0–2, mid 3–5; senior 6+ reference)			
Application, junior	2.13	3.95	11.36*
	(1.65)	(6.66)	(6.88)
Application, mid	−2.79**	−2.39	−10.09**
	(1.35)	(3.87)	(5.01)
Firms	643	109	35

Notes: Percentage points; robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Each entry is the firm-weighted average of (change in the share of the firm's AI jobs at the level, pre to post November 2022) minus (the same change for its non-AI technical and engineering jobs). Firms need at least five jobs in each group and period. The United Kingdom and the Philippines have too few qualifying firms (18 and 15) to report; Building cells outside the United States and India are too small.

None of the Indian estimates is distinguishable from zero, and the point estimates for junior Application jobs are positive: AI jobs' junior share did not fall by more than that of comparable non-AI jobs at the same firms. The comparison rests on 109 Indian firms and is imprecise, so it neither confirms nor rules out an AI-specific component; it does not support attributing the Indian decline to AI-related work.

D Data construction

Panel firms are mapped from `unique_firm_id` to LinkUp company identifiers through the firm lookup used throughout. The excluded firms are `firm_002877` (KPMG), `firm_003084` (Deloitte & Touche) and `firm_007589` (Infosys), identified by a whole-word name match checked by hand.

One pass over LinkUp's 291 raw `job_record` shards keeps every posting of panel firms in the five countries. Duplicate `JOB_HASH`es across shards are removed, keeping the earliest record. A distinct job is a unique (firm, normalized title, creation date); titles are normalized by lower-casing and replacing every non-alphanumeric character with a space. Location

copies differ by country and task: a U.S. Building job appears 3.07 times on average against 2.46 for a U.S. Application job, and an Indian Building job 1.63 times against 1.98. Raw posting counts would therefore overstate the U.S. share of Building work.

The location-cost terms are applied case-insensitively to U.S. descriptions. *Clearance:* a security, secret, government, federal or DoD clearance; an active, current, obtained, maintained or held (security) clearance; "clearance required"; top secret; TS/SCI; U.S. citizen or citizenship; ITAR; public trust. *On-site:* on-site, onsite, in-office, in person. *Client-facing:* client-facing, customer-facing, working directly with (our) clients or customers, client meetings or engagements, stakeholder management. *Travel:* travel. *Remote:* remote. A bare "clearance" pattern was dropped before estimation because it matched phrases such as "clearance of backlog". The exact expressions are in the replication package.

The skill-intensity terms are applied case-insensitively to U.S. descriptions of titles in the six-category and ordinary technical lookups. *PhD:* PhD, Ph.D., doctorate, doctoral degree. *Master's:* master's, masters degree, master of science, engineering, arts or computer, M.S. degree or M.S. in, MS degree, MS in a technical field, MSc, M.Tech ("MS" alone was dropped because it matches MS Office). *Research record:* publications in, publication record, published research or papers, peer-reviewed, first-author, top-tier journals, conferences or venues (a bare "publication" was dropped because it matched "DoD Publication 8570").

Each category's seed vocabulary (script `relabel_no_general_tech` in the replication code) is matched, token by token, against the twenty most distinctive terms of each group; a group needs at least one match. Abbreviated: *LLM_BUILDER:* PyTorch, TensorFlow, deep learning, neural networks, transformer architecture, LLM training, fine-tuning, pretraining, foundation model, distributed training, CUDA, machine learning engineer, research or applied scientist. *DATA_MLOPS_ENGINEER:* MLOps, Kubeflow, MLflow, model registry, model serving and monitoring, feature store, CI/CD and data pipelines, ETL, Airflow, Spark, Kafka, Kubernetes, Docker, Terraform, platform, site-reliability and DevOps engineer. *TECHNICAL_LLM_USER:* OpenAI and Anthropic APIs, LangChain, LlamaIndex, prompt engineering, retrieval-augmented generation, vector databases, semantic search, function calling, agentic workflows, API integration. *FUNCTIONAL_LLM_USER:* AI strategy, adoption, transformation and roadmap, digital transformation, AI product or program manager, change management, business case, stakeholder management, AI consultant. *AI_UX_INTERACTION:* conversational and interaction design, human–AI interaction, UX research, usability testing, human-centered design. *AI_GOVERNANCE_RISK:* AI governance, responsible AI, AI ethics and risk management, model risk management, algorithmic fairness, AI safety, regulatory compliance, data privacy, GDPR, model validation, privacy engineer.

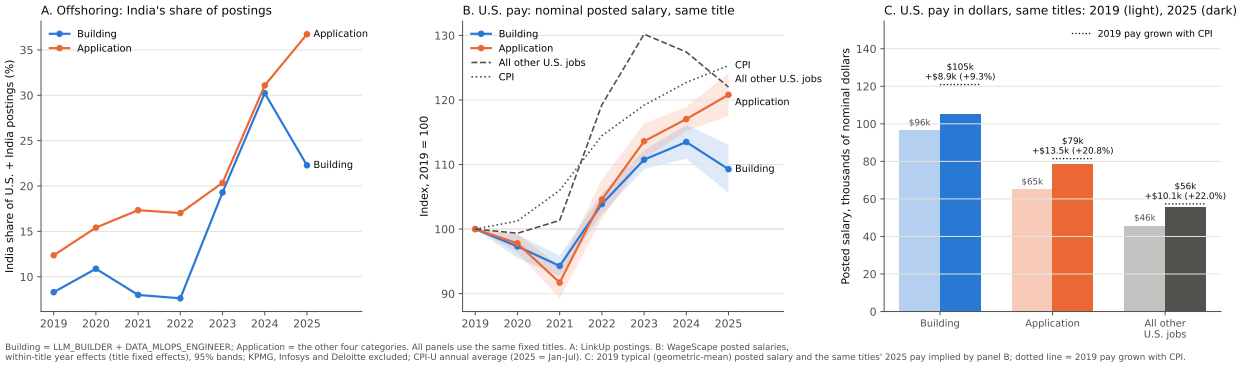
Table 18: Distinct focal-category jobs by year and country

Year	United States	India	United Kingdom	Mexico	Philippines	India share (%)
2019	102,588	18,197	10,046	1,762	1,095	15.1
2020	80,781	16,913	6,287	1,400	811	17.3
2021	144,535	38,267	13,950	4,072	2,155	20.9
2022	151,940	45,182	16,763	4,293	2,832	22.9
2023	99,817	33,046	14,416	3,816	2,815	24.9
2024	81,456	35,662	10,951	3,504	2,900	30.4
2025	71,056	38,342	8,469	3,362	2,828	35.0
Total	732,173	225,609	80,882	22,209	15,436	23.6

Notes: Distinct jobs in the six focal categories of the fixed title lookup. India share: India's percent of combined U.S. and Indian jobs. Year-to-year changes partly reflect LinkUp's scraping coverage.

Required experience is parsed with the same English pattern in every country (scripts `extract_experience_required` for the United States and India and `extract_experience_country` for the other countries, in the replication code). A Spanish variant adds under one percentage point of coverage in Mexico. Distinct jobs keep the minimum requirement across their copies.

E Additional wage figures (online appendix)



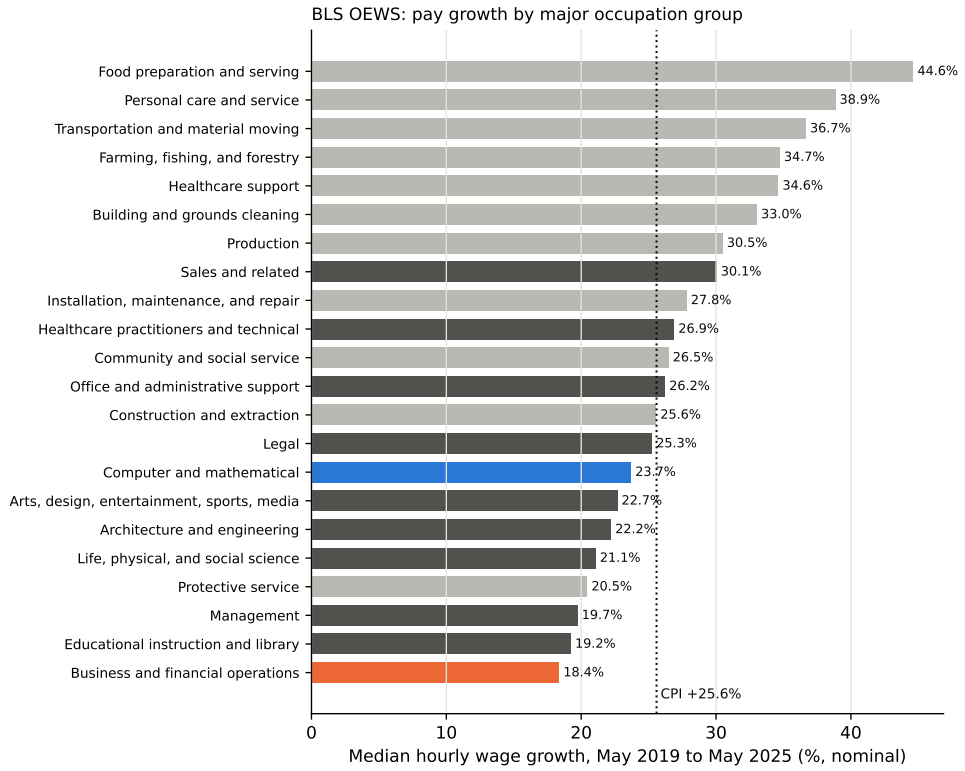
Building = LLM_BUILDER + DATA_MLOPS_ENGINEER; Application = the other four categories. All panels use the same fixed titles. A: LinkUp postings. B: WageScape posted salaries, within-title year effects (title fixed effects), 95% bands; KPMG, Infosys and Deloitte excluded; CPI-U annual average (2025 = Jan-Jul). C: 2019 typical (geometric-mean) posted salary and the same titles' 2025 pay implied by panel B; dotted line = 2019 pay grown with CPI.

Figure 11: Task offshoring and U.S. posted pay by year, 2019–2025. Panel A: India's share of U.S.-plus-India LinkUp postings in each task group, on a fixed set of job titles. Panel B: posted salaries in WageScape, from regressions of log salary on year indicators with job-title fixed effects, indexed to 2019 = 100 (95 percent bands, clustered by title); "All other U.S. jobs" is a 2 percent random sample of all other WageScape jobs; CPI-U is the annual average (2025 = January–July). Panel C: the same within-title change in dollars. KPMG, Deloitte and Infosys are excluded in all panels.

Table 19: Largest titles in each category, distinct jobs 2019–2025

Category	Title (normalized)	United States	India	Other three
LLM_BUILDER	network engineer	5,375	1,541	790
	machine learning engineer	1,865	587	241
	senior machine learning engineer	895	314	114
	network architect	617	173	127
	network operations engineer	435	103	172
	network technician	410	18	20
DATA_MLOPS_ENGINEER	cloud engineer	1,851	794	373
	cloud architect	996	506	196
	operations asm	964	2	11
	cloud software engineer	591	128	91
	lead devops engineer	386	405	210
	devops architect	378	331	27
TECHNICAL_LLM_USER	systems engineer	9,014	895	1,235
	senior systems engineer	4,677	608	622
	embedded software engineer	2,865	328	304
	software engineer iii	2,122	1,338	294
	sr systems engineer	1,750	294	57
	principal systems engineer	1,565	110	177
FUNCTIONAL_LLM_USER	project manager	27,610	6,030	9,061
	program manager	14,493	2,156	1,353
	business analyst	11,369	7,200	4,252
	assistant manager	9,271	4,728	999
	machine operator	8,801	36	287
	data analyst	6,585	3,393	2,356
AI_UX_INTERACTION	test engineer	4,857	1,714	1,332
	software test engineer	1,745	559	300
	qa engineer	1,509	1,893	544
	senior test engineer	1,347	1,035	425
	java full stack developer	891	1,103	55
	user experience designer	863	232	157
AI_GOVERNANCE_RISK	cyber security engineer	1,239	202	168
	cyber security analyst	1,096	577	233
	information security analyst	1,013	452	204
	security analyst	926	421	302
	cybersecurity engineer	786	204	61
	cybersecurity analyst	577	91	43
GENERAL_TECH	technical support specialist	2,014	374	470
	desktop support technician	1,567	61	79
	it support specialist	1,460	228	320
	it support technician	1,120	63	197
	support lead part time	1,083	0	53
	technical support representative	888	194	293

Notes: The six largest titles of each category in the fixed lookup, by U.S. distinct jobs. "Other three": United Kingdom, Mexico and the Philippines combined. Titles are shown as normalized (lower case, punctuation removed).



Blue: computer and mathematical. Orange: business and financial operations (closest to FUNCTIONAL_LLM_USER). Dark grey: other white-collar groups; light grey: other groups. Source: BLS OEWS national estimates, May 2019 and May 2025.

Figure 12: Median hourly wage growth by BLS major occupation group, May 2019 to May 2025, nominal. Dotted line: CPI-U over the same months. Source: BLS OEWS.

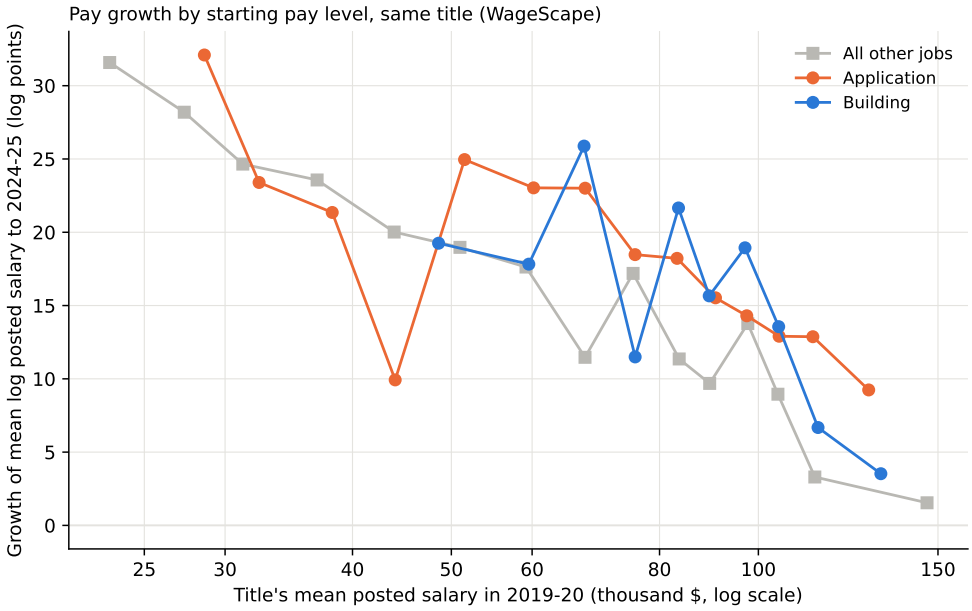


Figure 13: Posted-pay growth by starting pay, within title. Posting-weighted mean growth of log posted salary from 2019–20 to 2024–25 among titles in each of 15 quantile bins of 2019–20 pay; titles with at least five postings in each period. WageScape; KPMG, Deloitte and Infosys excluded.

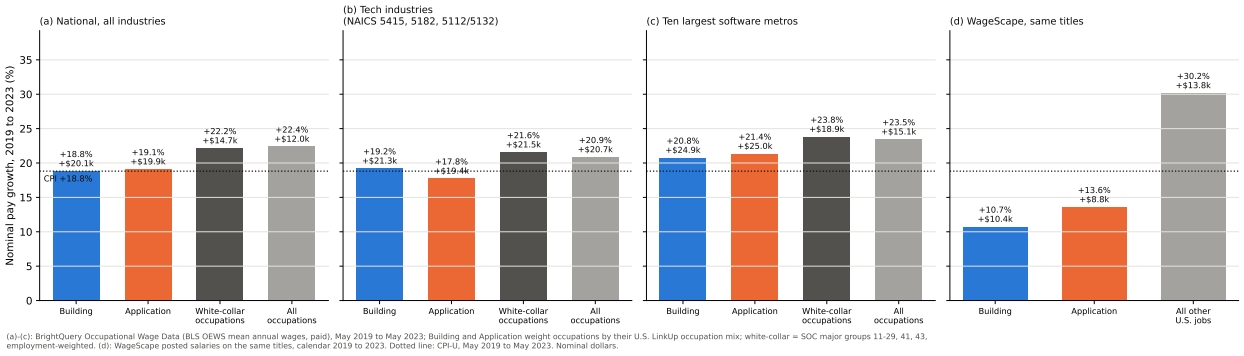


Figure 14: U.S. pay growth by task group, May 2019 to May 2023, in percent with the dollar change above each bar. Panels (a)–(c): BrightQuery Occupational Wage Data (BLS OEWS mean annual wages, paid), nationally, within NAICS 5415, 5182 and 5112/5132, and within the ten metropolitan areas with the most software developers in 2019; Building and Application weight occupations by their U.S. LinkUp occupation mix; white-collar is SOC major groups 11–29, 41 and 43. Panel (d): WageScape posted salaries on the same titles, 2019 to 2023. Dotted line: CPI-U. The window ends in 2023, the last year of the BrightQuery data; the comparison was specified before estimation (notebook `grh_pay_dollars_brightquery_wagescape`).

F Validating the seven categories against job titles and occupations (online appendix)

A useful category measure should agree with independent signals of the work where those signals are informative, and should add information where they are not. This appendix checks both for the full-text categories of Section 5.1, on the 9.1 million U.S. and Indian technical postings of 2019–2025 that carry one. The full-text encoder reads the title with each window of the description, so agreement with the title is partly mechanical; the title-blind known-title check of Section 5.1 (73.9 percent) is its complement.

Table 20, Panel A, compares the full-text category with two title-driven labels of the same posting: the topic-model category, and the fixed title lookup used for the five-country results. Agreement on the seven categories is 32 to 46 percent and on the three groups (Building, Application, general tech) 63 to 80 percent; Cohen's kappa is 0.21 to 0.30, fair agreement. The disagreement has one main source. The title-based labels place a large share of every category in functional application, the broadest category (it holds 63 to 79 percent of lookup jobs; Section 5.2): 48.5 percent of the U.S. postings that the full text reads as model building carry a lookup title in functional application, typically a generic analyst or manager title. Apart from that catch-all, each full-text category's most common lookup counterpart is its own category in the United States, and in India for every category except technical application, whose lookup titles split between technical application and product and testing (Panel B). The title-based categories are therefore a coarser version of the same classification, reliable for the Building–Application–general tech split and less so for the finer categories, which is why the U.S.–India results use the full text.

Panel C relates the full-text category to the posting's O*NET occupation. The two are related (Cramér's V of 0.50 in the United States and 0.48 in India), and they agree where they should: 86 percent of U.S. information-security analysts are in governance and security, 79 percent of computer user support specialists in general tech, and data scientists divide between model building (42 percent) and data and ML infrastructure (32 percent). But the occupation does not determine the category. Assigning each posting its occupation's most common category reproduces the full-text category for only 53 percent of U.S. and 59 percent of Indian postings, and in 73 and 83 percent of the occupations with at least 1,000 postings no category holds 80 percent of them. Software developers, the largest occupation, are 48.5 percent technical application in the United States and 65.4 percent in India, with the rest spread over model building, data infrastructure, AI product and testing and general tech. The categories carry information about the work that the occupation code does not.

Table 20: The full-text categories against title-based labels and O*NET occupations

A. Agreement of the full-text category with title-based labels of the same posting

Comparison	Country	Postings	Agree, 7 categories (%)	Agree, 3 groups (%)
Topic-model category	United States	6,153,604	32.2	63.0
	India	2,909,963	38.9	69.9
Fixed title lookup	United States	1,839,883	37.9	69.0
	India	426,471	45.6	79.5

B. Share of each full-text category whose lookup title is in the same category (%), and in functional application

	Same, U.S.	Same, India	Functional, U.S.
Model building	17.7	25.2	48.5
Data and ML infrastructure	28.1	30.4	37.7
Technical application	40.7	24.0	18.0
Functional application	87.9	96.8	—
product and testing	9.8	22.8	73.4
governance and security	18.7	31.8	54.7
General tech	27.4	26.7	35.8

*C. The full-text category and the O*NET occupation*

	United States	India
Cramér's V	0.50	0.48
Occupation's modal category correct (%)	52.6	58.7
Occupations with ≥1,000 postings	45	35
of which no category holds 80% (%)	73.3	82.9

Notes: U.S. and Indian technical postings of the panel firms with a full-text category, 2019–2025 (KPMG, Deloitte a Panel A: postings whose comparison label is one of the seven categories. Three groups: Building (model building, data an Application (the other four focal categories), general tech. Panel B: row percentages of the confusion matrix against the postings with an O*NET code; the modal-category accuracy assigns each posting the most common full-text category of it notebook `jmp_category_validation`.

G Measured location-cost attributes (online appendix)

The main text measures tasks by their category. This appendix reports a complementary measure: four requirements that U.S. employers write into their postings and that tie a task to the United States (security clearance or citizenship, client contact, on-site work, travel). Entered in Equation (5) as the attributes x_z, they give the destination-specific prices $\beta_d^\tau = -\theta\delta_d$ of Assumption 2 directly, and so a direct test of the single ranking of tasks in Grossman and Rossi-Hansberg (2008).

G.1 Measuring the cost of performing a task outside the United States

In the task-trade model, a task is offshored when the wage saving exceeds $\tau(z)$, the cost of performing it away from headquarters. The organizational and trade literatures identify the sources of that cost: the tacit knowledge and communication a task requires, its need for proximity to customers, its need for physical presence, and legal restrictions on where it may be performed (Antràs et al. 2006; Blinder and Krueger 2013). Each of these leaves a trace in what U.S. employers write into their postings, and I measure four of them directly.

- *Legal and security requirements.* A requirement of a security clearance, U.S. citizenship, ITAR compliance or public-trust eligibility means the task cannot legally be performed abroad. This is the purest form of location cost.
- *Client contact.* Client-facing work, direct work with customers, client meetings and stakeholder management require communication that is costly across distance and time zones, and proximity to the clients themselves.
- *Physical presence.* On-site, in-office or in-person requirements tie a task to a facility: hardware, laboratories, secure premises or teams that work face to face.
- *Travel.* A requirement to travel ties the task to a U.S. base from which clients or sites are visited.

For each title in the lookup I compute, from 2.07 million U.S. postings with full text, the share that mention each requirement (the terms were fixed before any estimation; Appendix D). On average 11.4 percent of a title's U.S. postings mention a clearance or citizenship requirement, 13.5 percent on-site work, 11.4 percent client contact and 41.1 percent travel. The composite *location-cost index* is the mean of these four standardized shares, minus the standardized share of postings advertised as remote, standardized across titles. The

components are nearly uncorrelated with one another ($|r| \leq 0.16$), so the index aggregates distinct sources of location cost rather than one source measured four times.

Two features of the construction matter for interpretation. Only U.S. text enters, so a destination's own postings—and the way they are written—cannot drive its score. And the index is defined for job titles, so it varies within occupations. Among the titles with the highest scores are cleared government and defense engineering roles and client-facing strategy roles; among the lowest are remote biostatistics, technical-support and product-management roles.

For comparison I also construct a Blinder–Krueger-style occupation index from O*NET Work Context: minus a standardized composite of face-to-face discussion, dealing with the public, physical proximity and public speaking, assigned to each title through its modal O*NET code.

G.2 Location costs and task location

Table 21 reports the central result. Within the same firm, title and year, a one-standard-deviation increase in location cost lowers the log odds that a job is placed in India rather than the United States by 0.26 (s.e. 0.05), or 23 percent in odds. The Philippines points the same way (-0.15, $p = 0.07$). Mexico and the United Kingdom show no gradient, which Section G.3 explains. Estimated for India against the United States alone, with firm × country, firm × decile, country × year and decile × year effects on deciles of the index, the gradient is -0.54 (s.e. 0.11). The relation does not reflect occupation mix. India's jobs score 0.032 standard deviations lower than the same firm's U.S. jobs in the same year ($t = -4.5$), and the difference is unchanged when O*NET occupation fixed effects are added.

Table 21: Location cost and task location

	India	Philippines	Mexico	United Kingdom
Destination × location cost (SD)	-0.258^{***}	-0.151^{*}	-0.035	0.031
	(0.051)	(0.082)	(0.072)	(0.049)
Firm × title × year FE	Yes			
Firm × country × year FE	Yes			
Cells used	1,013,952			

Notes: One PPML regression of Equation (9), the four destination interactions estimated jointly. Outcome: distinct jobs in firm × country × title × year cells, 2019–2025. Location cost: composite index from U.S. posting text (Section G.1), standardized across titles. Standard errors clustered by firm in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The occupation-level index fails the same test, and the reason is instructive. With the

Blinder–Krueger-style index, the within-firm gradient for India is +0.056 ($p = 0.04$); it falls to −0.013 (s.e. 0.058) once Software Developers, about 30 percent of jobs, are excluded. Personal contact varies little across technical occupations, and one occupation drives the index. The text-based measure succeeds because the relevant variation lies within occupations.

G.3 Destination-specific prices of the attributes

Table 22 enters the separate requirements jointly, and Figure 15 plots them. India avoids every form of U.S.-tied task: client-facing ($t = -3.5$), on-site ($t = -3.4$) and clearance-requiring work ($t = -2.2$). The Philippines avoids client-facing work. Mexico shows no aversion to on-site or client-facing work. The United Kingdom receives *more* client-facing ($t = 2.8$) and on-site work ($t = 3.2$), and less work requiring U.S. clearance or citizenship ($t = -3.3$).

Table 22: Which requirements keep work in the United States, by destination

	India	Philippines	Mexico	United Kingdom
Client-facing	−0.134***	−0.201***	−0.091	0.087***
	(0.039)	(0.071)	(0.062)	(0.031)
On-site	−0.122***	−0.057	0.002	0.097***
	(0.035)	(0.051)	(0.033)	(0.030)
Clearance / citizenship	−0.099**	−0.007	0.063	−0.224***
	(0.044)	(0.060)	(0.048)	(0.068)
Remote	−0.169**	−0.068	0.029	0.018
	(0.070)	(0.086)	(0.051)	(0.038)
Building (0/1)	−0.089	−0.571***	−0.434***	−0.099
	(0.098)	(0.144)	(0.112)	(0.076)

Notes: One PPML regression of Equation (9) with all 20 destination × attribute interactions. Requirements are title-level shares of U.S. postings, standardized; Building indicates LLM_BUILDER or DATA_MLOPS_ENGINEER titles. Fixed effects and sample as in Table 21. Standard errors clustered by firm in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The cross-destination contrasts follow the destinations' profiles closely. The United Kingdom receives 0.22 log points more client-facing work than India (s.e. 0.04, $t = 5.5$) and 0.22 more on-site work ($t = 6.3$), with similar gaps against the Philippines and Mexico: client-facing work goes where the clients are. Mexico receives 0.12 log points more on-site work than India ($t = 3.0$), consistent with nearness and shared working hours, though not significantly more client-facing work. Each destination thus receives the tasks whose location costs it is best placed to absorb. A model in which location cost belonged to the task alone could not generate this pattern.

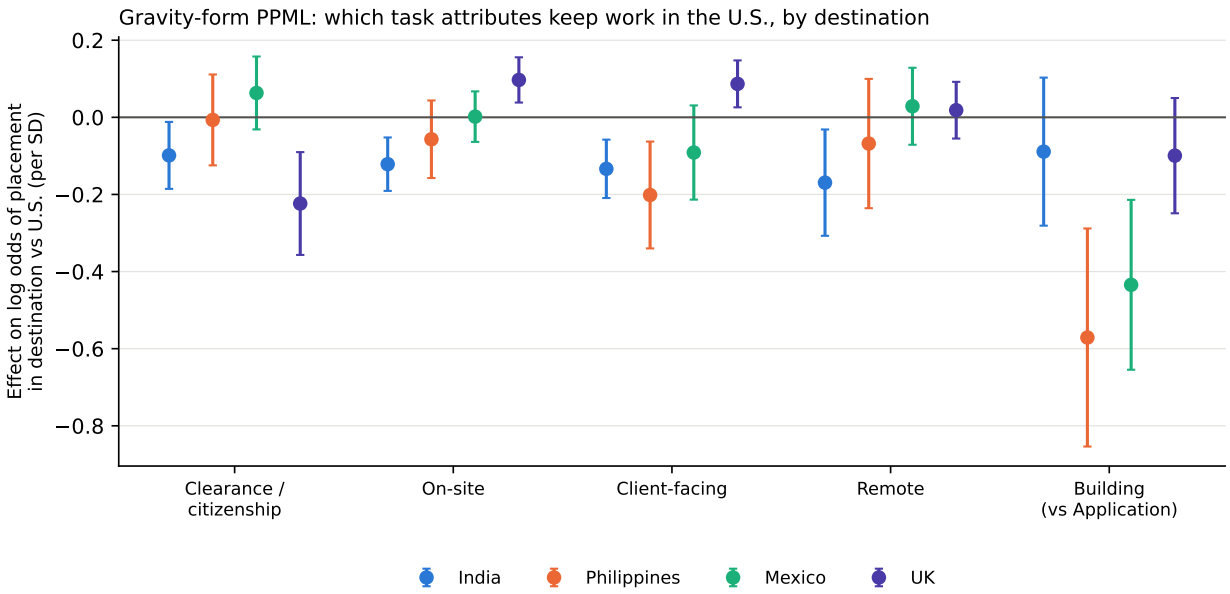


Figure 15: Effect of each requirement on the log odds of placing a job in the destination rather than the United States. Coefficients and 95 percent intervals from Table 22; requirements per standard deviation, Building as a 0/1 indicator.

One coefficient runs against intuition: India avoids titles that are frequently advertised as remote in the United States ($t = -2.4$). One reading is that U.S. firms fill remote-capable roles with remote domestic workers, a substitute for offshoring that the pandemic made available (Dingel and Neiman 2020). I note it but do not test it.

H Extension: the experience ladder within each country

Equation (6) holds senior hiring fixed. To describe the whole ladder, give each country's operation a three-layer knowledge hierarchy (Garicano and Rossi-Hansberg 2006). Juniors execute codified tasks; mid-level engineers execute and solve the problems within their knowledge K_c, a share $F(K_c)$; the rest are escalated to seniors, each taking h units of senior time. With ℓ_c juniors per mid-level engineer,

$$n_c^J : n_c^M : n_c^S \;=\; \ell_c \;:\; 1 \;:\; q_c, \qquad q_c \equiv h\,[1 - F(K_c)]. \tag{13}$$

In a team that spans countries (Antràs et al. 2006), problems that need tied or doctoral knowledge are escalated to the United States, so offshore seniors mostly resolve local exceptions. Generative AI lowers ℓ_c by performing codified tasks and lowers q_c by putting

knowledge within reach.

PROPOSITION 8 (The shape of the ladder). *Let AI lower ℓ_c by the proportion $\hat{\ell}_c$ and q_c by the proportion $\hat{q}_c$. (i) The mid-level share rises. (ii) If $\hat{q}_c = 0$, the junior share falls and the senior share rises. (iii) Junior and senior shares both fall—the ladder becomes a diamond—if and only if*

$$\frac{\ell_c}{1+\ell_c} \;<\; \frac{\hat{q}_c}{\hat{\ell}_c} \;<\; \frac{1+q_c}{q_c}.$$

Offshoring selection makes both margins large offshore: junior work there is mostly codified, and offshore seniors resolve problems that AI puts within reach of mid-level engineers. At home, seniors resolve frontier and doctoral problems that AI reaches least, so $\hat{q}_{US}$ is small and part (ii) applies. The proposition is static; Friebel et al. (2026) supply the dynamics, in which a productivity shock causes a temporary junior freeze and a learning shock a permanent change of shape (P6′).

I Proofs for the model

This appendix derives the results of Section 3 in the order in which they appear there.

Shares from job-level draws

By Equation (1), job j of task z at firm i costs $C_{ijzc} = \bar{C}_{izc}/\varepsilon_{ijc}$ in country c, with deterministic part

$$\bar{C}_{izc} \equiv a(z)\,\tau_c(z)\,(w_c^H)^{s_z}(w_c^L)^{1-s_z}\,\phi_{ic}.$$

For Lemma 1, read $\tau_c(z)(w_c^H)^{s_z}(w_c^L)^{1-s_z}$ as a single unit cost $w_c\,t_c(z)$; Assumptions 2 and 3 give it structure, and the derivation of the shares does not depend on that structure.

Under Assumption 1, for any $x > 0$,

$$\Pr(C_{ijzc} > x) = \Pr\big(\varepsilon_{ijc} < \bar{C}_{izc}/x\big) = \exp\big(-(x/\bar{C}_{izc})^{\theta}\big),$$

so the density of C_{ijzc} is $g_c(x) = \theta\,x^{\theta-1}\bar{C}_{izc}^{-\theta}\exp\big(-(x/\bar{C}_{izc})^{\theta}\big)$.

The job goes to c if C_{ijzc} is below every other country's cost. With draws independent across countries,

$$\pi_{izc} = \int_0^\infty g_c(x)\prod_{c'\neq c}\Pr(C_{ijzc'} > x)\,dx = \int_0^\infty \theta\,x^{\theta-1}\bar{C}_{izc}^{-\theta}\exp\Big(-x^{\theta}\textstyle\sum_{c'}\bar{C}_{izc'}^{-\theta}\Big)\,dx = \dfrac{\bar{C}_{izc}^{-\theta}}{\sum_{c'}\bar{C}_{izc'}^{-\theta}},$$

using $\int_0^\infty \theta x^{\theta-1} e^{-Kx^\theta} dx = 1/K$. The factor $a(z)^{-\theta}$ appears in the numerator and in every term of the denominator and cancels. With the structure of Steps 2 and 3, this is Equation (3).

As $\theta \to \infty$, $\pi_{izc} \to 1$ for the country with the lowest $\bar{C}_{izc}$ and 0 for the others. With one destination and $\bar{C}_{iz,F}/\bar{C}_{iz,US} = w_F\, b\, t(z)/w_{US}$, the task is offshored if and only if Equation (2) holds.

Location costs, skill costs and the comparison with the United States

Taking logs of $\bar{C}_{izc}$ and using Assumption 2 and $\rho_c = \log(w_c^H/w_c^L)$,

$$\log \bar{C}_{izc} = \log a(z) + \kappa_c + x_z'\delta_c + \log w_c^L + s_z\, \rho_c + \log \phi_{ic},$$

since $(w_c^H)^{s_z}(w_c^L)^{1-s_z} = w_c^L\, e^{s_z \rho_c}$. Assumption 3 is what allows s_z to carry no country subscript. Dividing π_{izd} by $\pi_{iz,US}$ removes the denominator of the share. With $\kappa_{US} = 0$ and $\delta_{US} = 0$,

$$\log \frac{\pi_{izd}}{\pi_{iz,US}} = -\theta\big(\log \bar{C}_{izd} - \log \bar{C}_{iz,US}\big) = \theta\Big[\underbrace{\log \tfrac{w_{US}^L \phi_{i,US}}{w_d^L \phi_{id}}}_{\omega_{id}} - \kappa_d - x_z'\delta_d - s_z(\rho_d - \rho_{US})\Big],$$

which is Equation (4). Assumption 4 states the premium as

$$\rho_c = \tfrac{1}{\sigma}\big(\log B_c - \log(H_c/L_c)\big), \tag{14}$$

and substituting it into $\beta_d^s = -\theta(\rho_d - \rho_{US})$ gives

$$\beta_d^s \;=\; \frac{\theta}{\sigma}\Big[\log \frac{H_d}{L_d} - \log \frac{H_{US}}{L_{US}}\Big] \;-\; \frac{\theta}{\sigma}\big[\log B_d - \log B_{US}\big]. \tag{15}$$

Proof of Proposition 2. From Equation (4), $\partial \log(\pi_{izd}/\pi_{iz,US})/\partial s_z = -\theta(\rho_d - \rho_{US})$, which is negative if and only if $\rho_d > \rho_{US}$. By Equation (14) with $B_d = B_{US}$, $\rho_d - \rho_{US} = \sigma^{-1}[\log(H_{US}/L_{US}) - \log(H_d/L_d)]$, positive if and only if $H_d/L_d < H_{US}/L_{US}$. This proves (i). For (ii), Equation (15) with $B_d = B_{d'}$ gives $\beta_d^s - \beta_{d'}^s = (\theta/\sigma)[\log(H_d/L_d) - \log(H_{d'}/L_{d'})]$, which has the sign of $H_d/L_d - H_{d'}/L_{d'}$. □

Because $\theta/\sigma > 0$ multiplies every destination's coefficient, the ordering in (ii) does not depend on the values of θ and σ, nor on a common attenuation from measurement error in s_z. It does depend on the demand shifters B_d; a lower B_d raises β_d^s.

The gravity equation and its estimation

Proof of Proposition 1. With N_{izt} jobs of task z placed independently, the number placed in c is multinomial with mean $N_{izt}\pi_{izct}$. LinkUp observes each posting in c with probability η_{ict}, independently of the task, so the observed count V_{iczt} has mean $\eta_{ict}N_{izt}\pi_{izct}$. Write $\Phi_{izt} \equiv \sum_{c'} \bar{C}_{izc't}^{-\theta}$. Taking logs and substituting $\log \bar{C}_{izc}$,

$$\log \mathbb{E}[V_{iczt}] = \underbrace{\log N_{izt} - \log \Phi_{izt} - \theta \log a(z)}_{\alpha_{izt}} \\ \underbrace{- \theta \left(\log w_{ct}^{L} + \kappa_{ct} + \log \phi_{ict}\right) + \log \eta_{ict}}_{\gamma_{ict}} + x_z' \underbrace{(-\theta\, \delta_c)}_{\beta_c^{\tau}} + s_z \big(- \theta \rho_{ct} \big).$$

The last term varies by country, year and task, so neither set of fixed effects absorbs it. Adding and subtracting $\theta s_z \rho_{US,t}$, the piece $-\theta s_z \rho_{US,t}$ varies only by task and year and moves into α_{izt}, leaving $s_z \beta_{ct}^{s}$ with $\beta_{ct}^{s} = -\theta(\rho_{ct} - \rho_{US,t})$ and $\beta_{US,t}^{s} = 0$. This is Equation (5). □

The step requires that LinkUp's coverage of a firm in a country does not differ across the firm's tasks within a year (Section 4.2). Because s_z is measured with error (Section 5.4), the estimated β_d^s are attenuated toward zero by the same factor in every destination, which leaves their ordering intact.

Conditional on the total $\sum_c V_{iczt}$, the counts are multinomial with the shares of Lemma 1. A Poisson likelihood with a fixed effect for each firm × task × year has the same first-order conditions for the slope coefficients as this multinomial likelihood, so Poisson pseudo-maximum likelihood with these fixed effects estimates β_c^{τ} and β_c^s consistently, and it remains consistent if the counts are not Poisson, provided the conditional mean is correct (Santos Silva and Tenreyro 2006). Firm–task–year cells in which the firm posts in one country only contribute zeros for the other countries, as the model implies they should.

Migration

Proof of Proposition 3. With $H_d = (1 - m_d)\bar{H}_d$ and $H_{US} = \bar{H}_{US} + \sum_{d'} m_{d'} \bar{H}_{d'}$, Equation (15) gives $\beta_d^s = (\theta/\sigma)[\log(1 - m_d) + \log \bar{H}_d - \log L_d - \log H_{US} + \log L_{US}] + \text{const}$. Differentiating,

$$\frac{\partial \beta_d^s}{\partial m_d} = -\frac{\theta}{\sigma}\Big[\frac{1}{1 - m_d} + \frac{\bar{H}_d}{H_{US}}\Big] < 0.$$

The first term is the loss of graduates in the destination and the second the gain at home. Emigration from d also lowers $\beta_{d'}^s$ for every other destination d', through the second term alone, by $(\theta/\sigma)\bar{H}_d/H_{US}$. □

The Building–Application gap

For a Building task z_B and an Application task z_A, applying Equation (4) to each and differencing gives

$$\log \frac{\pi_{iz_B dt}}{\pi_{iz_B,US,t}} - \log \frac{\pi_{iz_A dt}}{\pi_{iz_A,US,t}} = -\theta\,(x_{z_B} - x_{z_A})'\delta_d \;-\; \theta\,(s_{z_B} - s_{z_A})(\rho_{dt} - \rho_{US,t}) \;-\; \theta\,\delta^B_{dt}, \quad (16)$$

where δ^B_{dt} is the Building-specific location cost that the measured attributes miss, added to $\log \tau_d(z_B)$.

Proof of Proposition 4. The terms ω_{id} and κ_d are the same for both tasks and cancel. For (i), the skill term $-\theta(s_{z_B} - s_{z_A})(\rho_{dt} - \rho_{US,t})$ is, by the argument of Proposition 2, most negative where H_d/L_d is smallest and zero where $H_d/L_d = H_{US}/L_{US}$. For (ii), when $s_z\beta^s_d$ is included in the regression, the destination × Building coefficient estimates $-\theta[(x_{z_B} - x_{z_A})'\delta_d + \delta^B_{dt}]$ only. For (iii), with time subscripts, the right-hand side of Equation (16) changes only through δ^B_{dt} or $\rho_{dt} - \rho_{US,t}$; κ_{dt} does not appear. □

The productivity effect

Proof of Proposition 6. The cost of job j in country c satisfies $\Pr(C_{ijzc} > x) = \exp(-(x/\bar{C}_{izc})^\theta)$, independently across countries, so the lowest cost $C^{\min}_{ijz}$ satisfies $\Pr(C^{\min}_{ijz} > x) = \exp(-x^\theta \Phi_{iz})$. This is a Weibull distribution with mean $\Gamma(1{+}1/\theta)\,\Phi_{iz}^{-1/\theta}$. From Lemma 1, $\Phi_{iz} = \bar{C}^{-\theta}_{iz,US}/\pi_{iz,US}$, so the mean equals $\Gamma(1 + 1/\theta)\,\bar{C}_{iz,US}\,\pi^{1/\theta}_{iz,US}$: the cost of doing the job at home, scaled down by the home share. Write the U.S. cost of task z in group g as $\bar{C}_{iz,US} = a(z)\,w_g$, where w_g is the group's U.S. wage (the U.S. wage index of Equation (1) for a group whose workers are paid alike). With Cobb–Douglas cost shares λ_z and zero profit,

$$\log p_g = \sum_{z \in g} \lambda_z \Big[\log \Gamma(1 + 1/\theta) + \log a(z) + \log w_g + \tfrac{1}{\theta} \log \pi_{z,US} \Big].$$

Differencing over time with $a(z)$ fixed and using $\sum_z \lambda_z = 1$ gives Equation (7). □

Proof of Proposition 7. By the argument for Proposition 6, with a common wage w_W the unit cost of task group g satisfies $\log c_g = \sum_{z \in g} \lambda_z \big[\log \Gamma(1{+}1/\theta) + \log a(z) + \frac{1}{\theta} \log \pi_{z,US} \big] + \log w_W$. With Cobb–Douglas shares μ_g and zero profit in white-collar services, $\log p = \sum_g \mu_g \log c_g +$ const. Differencing with $a(z)$ fixed and using $\sum_g \mu_g = \sum_{z \in g} \lambda_z = 1$ gives Equation (8). Part (i): with $\Delta \log p = 0$, every term $-\Delta \log \pi_{z,US} \geq 0$ and at least one is positive, so $\Delta \log w_W > 0$, and w_W is the wage in every group. Part (ii) follows because there is one wage. Part (iii) rearranges Equation (8). The home shares $\pi_{z,US}$ are themselves functions of w_W; the equation holds for their equilibrium values, which is what the data record. □

The two channels for generative AI

Proof of Proposition 5. Junior hiring in country c is $J_c = \sum_z a^J(z)\, N_z \pi_{zc}$, summing over firms implicitly.

(i) By Equation (4), lowering $\log \tau_d(z)$ by ξ_{dt} for every task above support raises their log odds in d, relative to support, by $\theta \xi_{dt}$. From Lemma 1, $\partial \log \pi_{zd} / \partial \log \tau_d(z) = -\theta(1-\pi_{zd}) < 0$ and $\partial \log \pi_{z,US} / \partial \log \tau_d(z) = \theta \pi_{zd} > 0$, so for every such task π_{zd} rises and $\pi_{z,US}$ falls; holding N_z fixed, J_d rises and J_{US} falls. A fall common to every task, support included, is a fall in κ_{dt}, which by Proposition 1 enters only γ_{ict}.

(ii) If AI lowers $a^J(z)$ by a factor $1 - \psi_z$ in every country, the cost of a job of task z in country c falls by the proportion $\psi_z \iota_{zc}$, where ι_{zc} is the junior share of its wage bill there. Under Assumption 7, $\iota_{zc} = \iota_z$, so the deterministic cost $\bar{C}_{izc}$ is multiplied by the same factor in every c, which cancels from π_{zc} as $a(z)$ does in Lemma 1. Then $J'_c = \sum_z (1-\psi_z) a^J(z) N_z \pi_{zc}$ and, to first order, $\Delta \log J_c \approx -\sum_z \nu_{zc} \psi_z$ with $\nu_{zc} = a^J(z) N_z \pi_{zc} / J_c$. Senior hiring is unchanged, so the junior share of hiring, ς_c^J, changes by approximately $(1 - \varsigma_c^J)\, \Delta \log J_c$ in logs. □

The shape of the ladder

In each country a mid-level engineer is matched with ℓ_c juniors who execute codified tasks. The mid-level engineer draws problems and solves those in the set of knowledge K_c within reach, a share $F(K_c)$. The rest are escalated, and each takes a senior h units of time, so one mid-level engineer generates demand for $q_c = h[1 - F(K_c)]$ seniors (Garicano and Rossi-Hansberg 2006). This gives Equation (13). Write $D_c = \ell_c + 1 + q_c$, so the shares of the three rungs are ℓ_c / D_c, $1/D_c$ and q_c / D_c.

Proof of Proposition 8. Let $d \log \ell_c = -\hat{\ell}_c \leq 0$ and $d \log q_c = -\hat{q}_c \leq 0$, not both zero. Then $d \log D_c = -(\ell_c \hat{\ell}_c + q_c \hat{q}_c)/D_c < 0$, so the mid-level share $1/D_c$ rises, which is (i). The junior share changes by

$$d \log \frac{\ell_c}{D_c} = -\hat{\ell}_c + \frac{\ell_c \hat{\ell}_c + q_c \hat{q}_c}{D_c} = \frac{-\hat{\ell}_c (1 + q_c) + q_c \hat{q}_c}{D_c},$$

which is negative if and only if $\hat{q}_c / \hat{\ell}_c < (1 + q_c)/q_c$. The senior share changes by

$$d \log \frac{q_c}{D_c} = -\hat{q}_c + \frac{\ell_c \hat{\ell}_c + q_c \hat{q}_c}{D_c} = \frac{-\hat{q}_c (1 + \ell_c) + \ell_c \hat{\ell}_c}{D_c},$$

which is negative if and only if $\hat{q}_c / \hat{\ell}_c > \ell_c / (1 + \ell_c)$. With $\hat{q}_c = 0$ the junior share falls and the senior share rises, which is (ii); both shares fall exactly when both inequalities hold, which

is (iii). □

In a team that spans countries (Antràs et al. 2006), problems that need tied or doctoral knowledge are escalated to U.S. seniors. Offshore seniors therefore handle the problems that are hard for the local team but not frontier problems, and these are the problems that a tool giving access to codified knowledge is most likely to put within reach: $F'(K_d)$ is large over the relevant range and $\hat{q}_d$ is large. U.S. seniors handle the frontier and doctoral problems, for which $\hat{q}_{US}$ is small, so the U.S. ladder is in case (ii). This is an assumption about where AI's knowledge is useful, not a result; the data in Table 14 are consistent with it.

Proposition 8 compares steady states. In the dynamic model of Friebel et al. (2026), juniors become seniors with some probability, so cutting junior hiring today lowers the supply of seniors later. A productivity shock then leads to a short-run freeze in junior hiring, after which the firm restores its long-run span, possibly after oscillating; a shock to the speed of learning changes the long-run span. A fall in q_c plays the role of their learning shock, since it lets mid-level engineers resolve what used to require seniority. I use their dynamic result as given and do not derive it here.

J Firms whose product is AI

The firms studied so far are, for the most part, users of AI: banks, retailers, consultancies and technology companies whose AI work is one activity among many. The sample also contains a small group of firms whose product *is* AI. I group twenty of them by business model, before looking at their postings: three model developers (OpenAI, Anthropic and DeepMind); six providers of AI compute and infrastructure (Nvidia, Cerebras, SambaNova, Graphcore, Anyscale and Databricks); and eleven AI software and platform firms (DataRobot, H2O.ai, Glean, SoundHound, Moveworks, Builder.ai, Instabase, Uniphore, Eightfold, C3 and Tiger Analytics). Several frontier developers that appear in LinkUp, among them Scale AI, Cohere, Mistral, Perplexity and xAI, are not in the sample, because it is restricted to firms advertising in both the United States and India.

These firms post few jobs, and many of their titles are not in the fixed lookup of Section 5.2. This section therefore uses every posting of the firms, in all titles, and reads each posting's duties directly. A posting counts as *model building* when its duties involve training, designing or substantively changing models; as *AI infrastructure* when they involve data and compute systems built to serve models; and as *AI application* when they involve integrating, deploying or using existing AI systems. The rules are applied sentence by sentence and ignore company boilerplate and lists of desired skills. They were fixed before any estimate was computed, but they have not yet been validated against human coding, so the estimates

in this section are provisional. Appendix K describes the human validation, whose sample has been drawn and whose coding is pending. Counts are *requisitions*: one advertisement text per firm and country, with the same text posted again after more than 30 days counted as a new requisition. This removes the many U.S. postings that repeat one opening across cities.

These firms advertise AI work of a different kind. Model building accounts for 5.8 percent of their requisitions and AI infrastructure for 20.3 percent, against 0.4 and 0.6 percent at the other firms. Their footprint abroad is small and recent. OpenAI's and Anthropic's first Indian advertisements appear in 2025 and DeepMind's in 2024, while the United Kingdom hosts a large share of DeepMind's and Graphcore's model-building requisitions. Because LinkUp began collecting some of these firms' sites only in 2023, entry dates partly reflect the start of collection rather than of hiring.

Table 23 estimates Equation (9) with the three task types as the task dimension, separately for these firms. The firm × task × year and firm × country × year effects play the same roles as in Section 7, and the coefficient of interest is the difference between a destination's coefficient on model building and on AI application. Unlike the average multinational, AI-native firms advertise model building outside the United States less than AI application, in India as well as in the United Kingdom. Within firm and year, the destination-to-U.S. ratio for model building is 31 percent lower than for application in India (-0.38 log points) and 46 percent lower in the United Kingdom (-0.61). A second definition, the 27 firms with at least 500 postings and at least 40 percent AI work among their 2019–2021 postings, gives similar estimates (-0.34 and -0.43). For all other firms the same contrast is close to zero in India (0.06, s.e. 0.14) and -0.26 (s.e. 0.07) in the United Kingdom, in line with the small Building–Application differences of Table 6. The twenty firms are too few to move these estimates.

The two results differ in strength. With twenty firms, conventional clustered standard errors are too small, so Table 23 also reports the range of estimates when each firm is dropped in turn and a bootstrap over firms. The United Kingdom estimate survives the exclusion of any single firm (range -0.69 to -0.47), and the bootstrap interval excludes zero (-0.95 to -0.11). The India estimate is negative whichever firm is excluded, but it is -0.22 without Nvidia, the largest advertiser, and its bootstrap interval (-0.57 to 0.12) includes zero. The India evidence is therefore suggestive. It is consistent with firms that develop models keeping that work close to their U.S. research organisations while placing application and integration work abroad, a pattern that does not appear among the much larger group of firms that use AI. The Mexico estimate rests on one or two firms and separates when C3 is dropped, so it is not informative. The Philippines is omitted because the twenty firms post no model-building

Table 23: AI-native firms: model building versus AI application, destination relative to the U.S.

	India	United Kingdom	Mexico
Named AI-native firms (20)	-0.38^{***}	-0.61^{***}	-0.99^{***}
	(0.08)	(0.17)	(0.21)
Pre-period AI-intensive firms (27)	-0.34^{***}	-0.43^{***}	-0.49^{**}
	(0.13)	(0.13)	(0.22)
All other firms	0.06	-0.26^{***}	-0.27^{*}
	(0.14)	(0.07)	(0.14)
All firms excluding the named AI-native firms	0.06	-0.26^{***}	-0.27^{*}
	(0.14)	(0.07)	(0.14)
Named firms: leave one firm out, range	[-0.41, -0.22]	[-0.69, -0.47]	[-13.40, -0.91]
Named firms: firm bootstrap 95% interval	[-0.57, 0.12]	[-0.95, -0.11]	[-14.61, -0.79]

Notes: $\beta_{dA} - \beta_{dC}$ (model building minus AI application) from PPML of requisition counts in firm × country × task × year cells with firm × task × year and firm × country × year fixed effects, estimated separately for each firm group; firm-clustered standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Named AI-native firms: 20 companies whose product is AI models, AI compute or AI software (OpenAI, Anthropic, DeepMind, Nvidia, Cerebras, SambaNova, Graphcore, Anyscale, Databricks, DataRobot, H2O.ai, Glean, SoundHound, Moveworks, Builder.ai, Instabase, Uniphore, Eightfold, C3, Tiger Analytics). Pre-period AI-intensive: 27 firms with at least 500 postings and at least 40% AI work among 2019–2021 postings. With 20 firms, clustered standard errors are unreliable; the bootstrap resamples firms (499 draws). The Mexico estimate rests on very few firms and separates when C3 is dropped. The Philippines is omitted: the named firms post no model-building requisitions there. Task labels are provisional (unvalidated description classifier).

requisitions there.

Figure 16 shows India's share of U.S.-plus-India requisitions by year. India's share of the AI-native firms' model-building requisitions rose to 28 percent in 2023 and then fell to 8 percent by 2025, as their U.S. model-building requisitions roughly tripled (from 139 to 408). Among the other firms, India's share kept rising, to 35 percent. The decline comes from growth in U.S. model building, not from fewer Indian requisitions (36 in 2025, against 54 in 2023). OpenAI's and Anthropic's first Indian advertisements (8 and 4 requisitions over the whole period) include no model building. The annual counts for AI-native firms are small.

These comparisons describe where firms advertise, not where they employ people, and they do not identify why AI-native firms differ. Their research staff, compute, data and leadership are concentrated in the United States, and any of these could produce the pattern. So could the fact that their Indian operations are new.

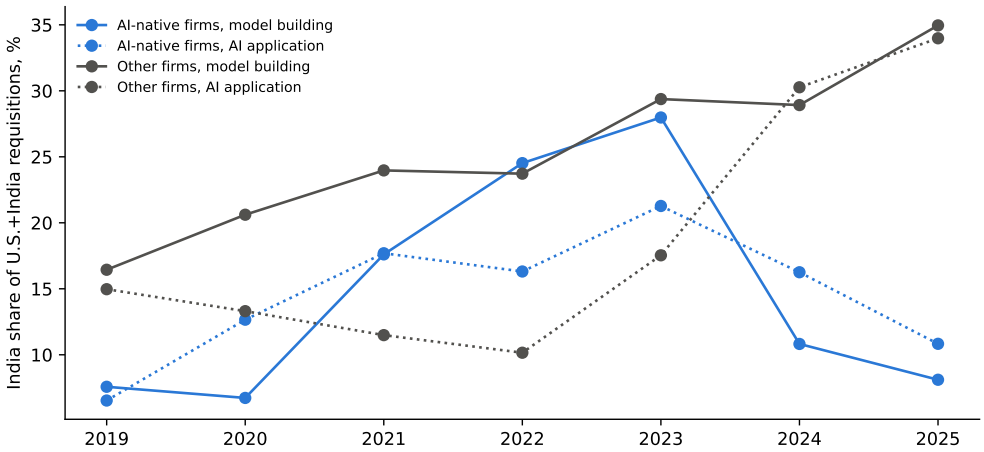


Figure 16: India's share of U.S.-plus-India AI requisitions, AI-native and other firms, by year, for model building (solid) and AI application (dotted). Requisitions labelled from posting descriptions (provisional). Counts for AI-native firms are small: 5 to 64 Indian model-building requisitions per year.

K Human validation of the task labels

The description-based task labels used in Section J come from a rule-based classifier. This appendix describes how they will be checked against human coding. The design was fixed and the sample drawn before any human label was seen. Coding is pending, so no accuracy figures are reported in this version.

Coders assign each advertisement to the categories its duties describe: model building (A), AI infrastructure (B), AI application (C) or non-AI technical work (D), ticking every category that applies, or mark it ambiguous or non-technical (E). They then choose a primary label: among A–C, the category that holds most of the job's AI duties, otherwise D. The codebook settles the hard boundaries in advance. "Network" means computer networking unless it is "neural network." Generic cloud or data-pipeline work is infrastructure only when its stated purpose is serving models. Fine-tuning is model building only when the posting describes retraining with data, weights or training methods. Software testing is not model evaluation. An AI term counts only when it describes the job's own duties, not the employer or a list of desired skills.

Table 24 describes the sample of 3,558 advertisements. Items are drawn within strata of country × period (before or from 30 November 2022) × classifier label, with at most two items per firm in each stratum. One part comes from postings in the fixed title lookup, and a second part, of 1,200 items, from all other postings, where most AI work lies. Further strata oversample the hard cases: network titles in the former model-building category, titles whose category the classifier disputes, AI terms that appear only incidentally or only in skills lists,

fine-tuning, software testing that mentions AI, and Spanish-language postings. 2,155 items are held out and will be used only to evaluate the classifier as frozen. The remaining items may be used to revise it, and any revised version will be evaluated on a fresh held-out draw.

Coders see the title and the full description only, not the country, the employer's category, the classifier's label or the reason an item was sampled. No posting text leaves the secure computing environment. Before coding, two coders label a common training set of 40 postings and discuss disagreements. A random quarter of the sample (919 items) is coded by two people. Agreement (percent agreement and Cohen's kappa) is reported before adjudication, and a third person resolves disagreements without seeing the classifier's label.

For held-out items, and by country and period, the next version will report the following:

- precision and recall of each label, weighted to the population of advertisements, with bootstrap intervals;
- confusion matrices, accuracy in each boundary stratum and the share of postings coders find ambiguous;
- the same statistics for the title categories used in Sections 5–7;
- the building-versus-application contrast after correcting the counts for each country's estimated misclassification.

The central question for this paper is whether the classifier errs differently in India than elsewhere, for example if Indian postings describe model building in vocabulary the rules miss. The country stratification is designed to answer it.

A second, smaller sample checks the full-text categories of Section 5.1, on the boundaries the known-title check flagged. It has 720 postings, 360 from each country. In each country, 120 are postings that the full-text measure places in model building, data and ML infrastructure or general tech but that the other text measures (a smaller embedding model on the full text, and the same model on the first part of the posting) place elsewhere; 120 are postings placed in technical or functional application with the same disagreement; and 120 are drawn at random. Coders see the title and description only and answer two questions: the category the duties describe, and whether the job is Building, Application or support work. The sample was drawn on 2026-10-15 under the same pre-specification as the estimates of Sections 7.1 and 7.1.

Table 24: The human validation sample

	Model building	AI infra.	AI application	Non-AI tech.	Ambiguous	Items	Held out	Double-coded	Boundary cases
United States	195	152	233	316	220	1116	696	269	379
India	155	144	157	141	91	688	405	192	84
United Kingdom	144	140	145	97	90	616	370	144	23
Mexico	130	122	142	98	109	601	370	165	47
Philippines	114	116	141	84	82	537	314	149	4
All	738	674	818	736	592	3558	2155	919	537

Notes: Blinded validation sample of job advertisements, by country and by the classifier's label. Items are drawn within strata of country × period (before / from 30 November 2022) × classifier label, with at most two items per firm per stratum, from the lookup-title postings and, separately, from all other postings. Boundary cases oversample hard categories: network titles in the former model-building category, title categories the classifier disagrees with, AI terms that appear only incidentally or only in skills lists, fine-tuning, testing with AI terms and Spanish-language postings. Held-out items are used only to evaluate the frozen classifier. Coders see the title and description only.